

Smart Digital Entry and Exit Management System for Residential Societies

Course Instructor: Ms. Laiba Imran

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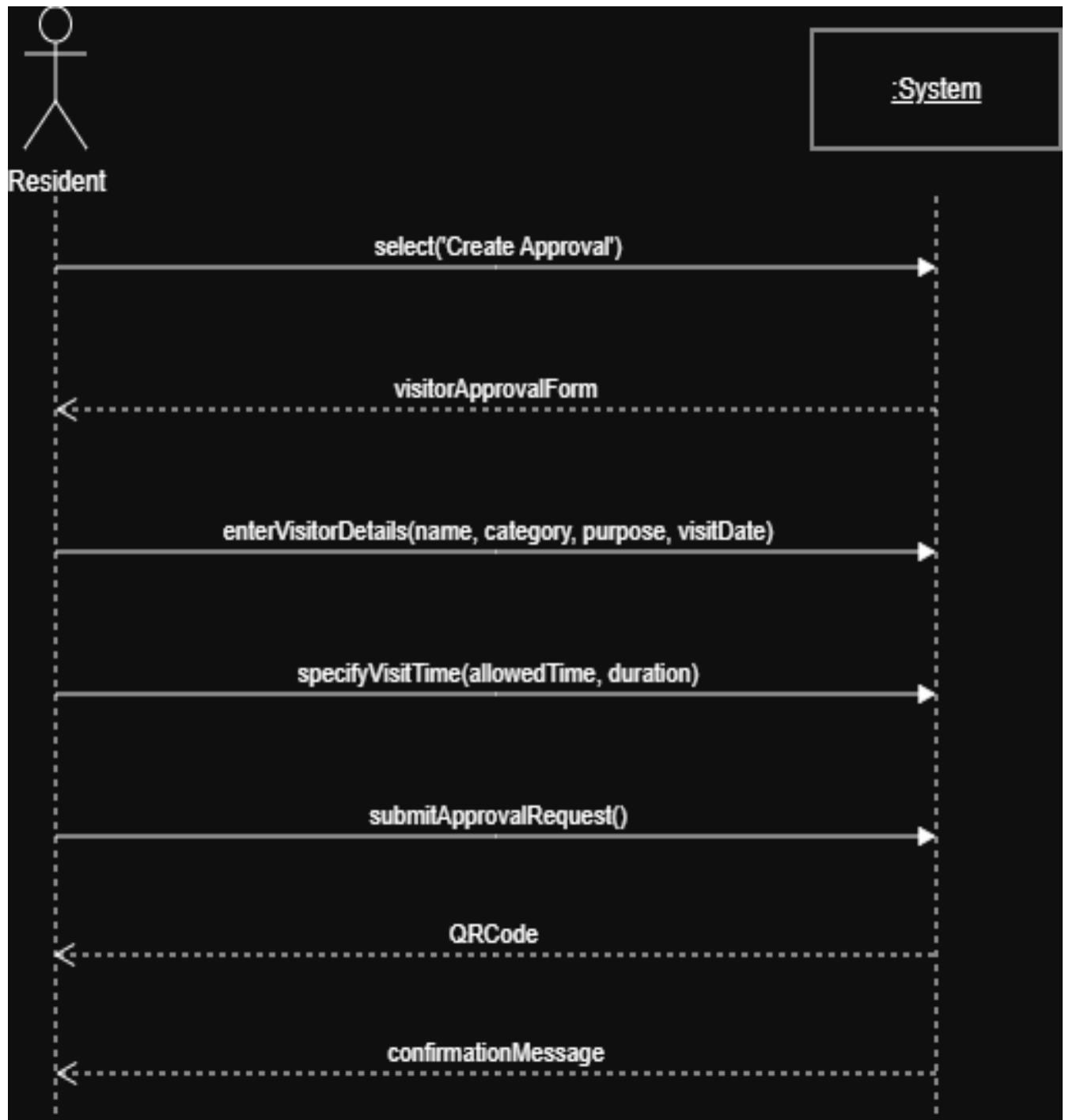


**Department of Computer Science
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Emerging Sciences**

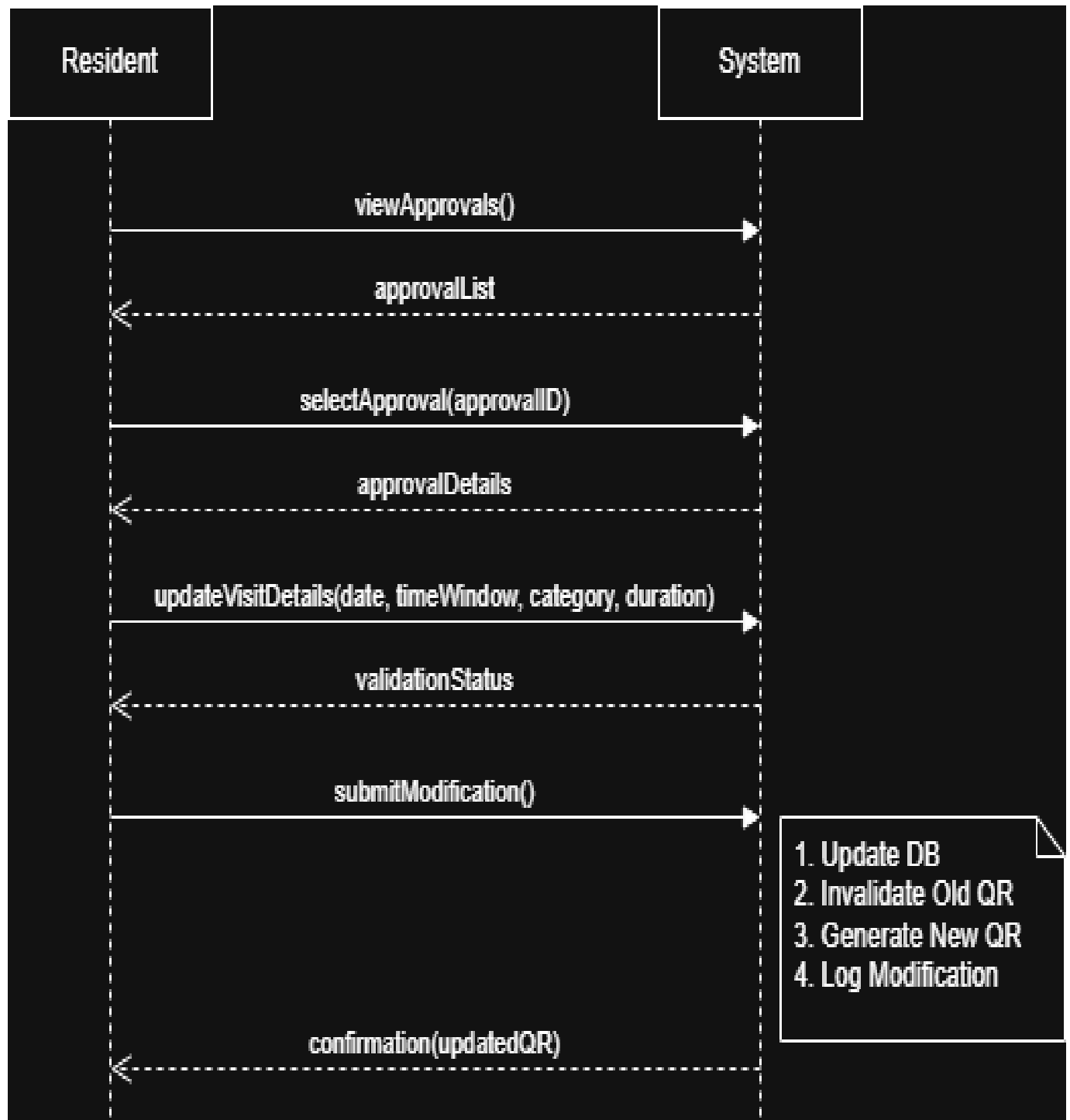
System Sequence Diagram:

Main Scenarios:

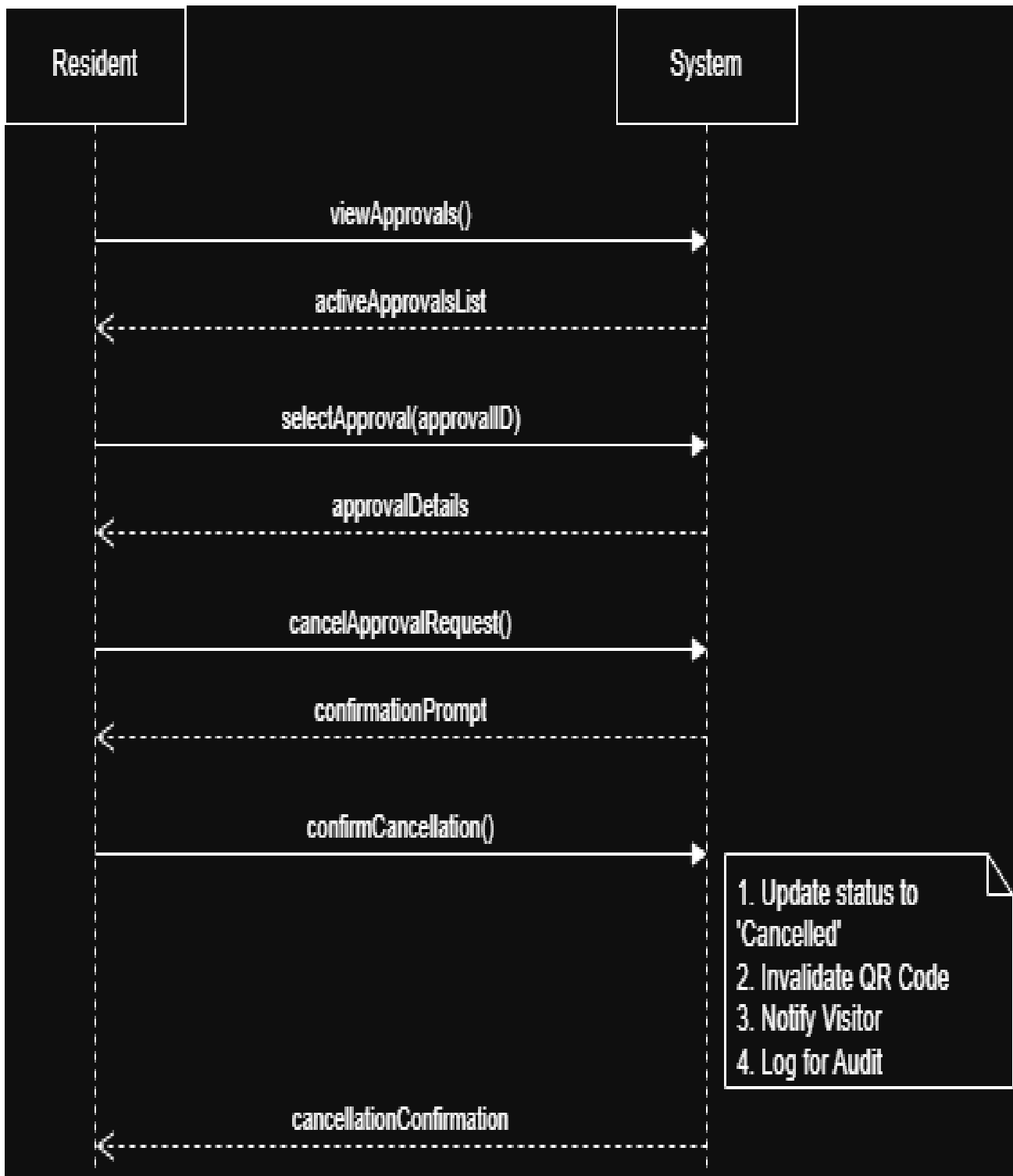
Use Case 1: Create approval



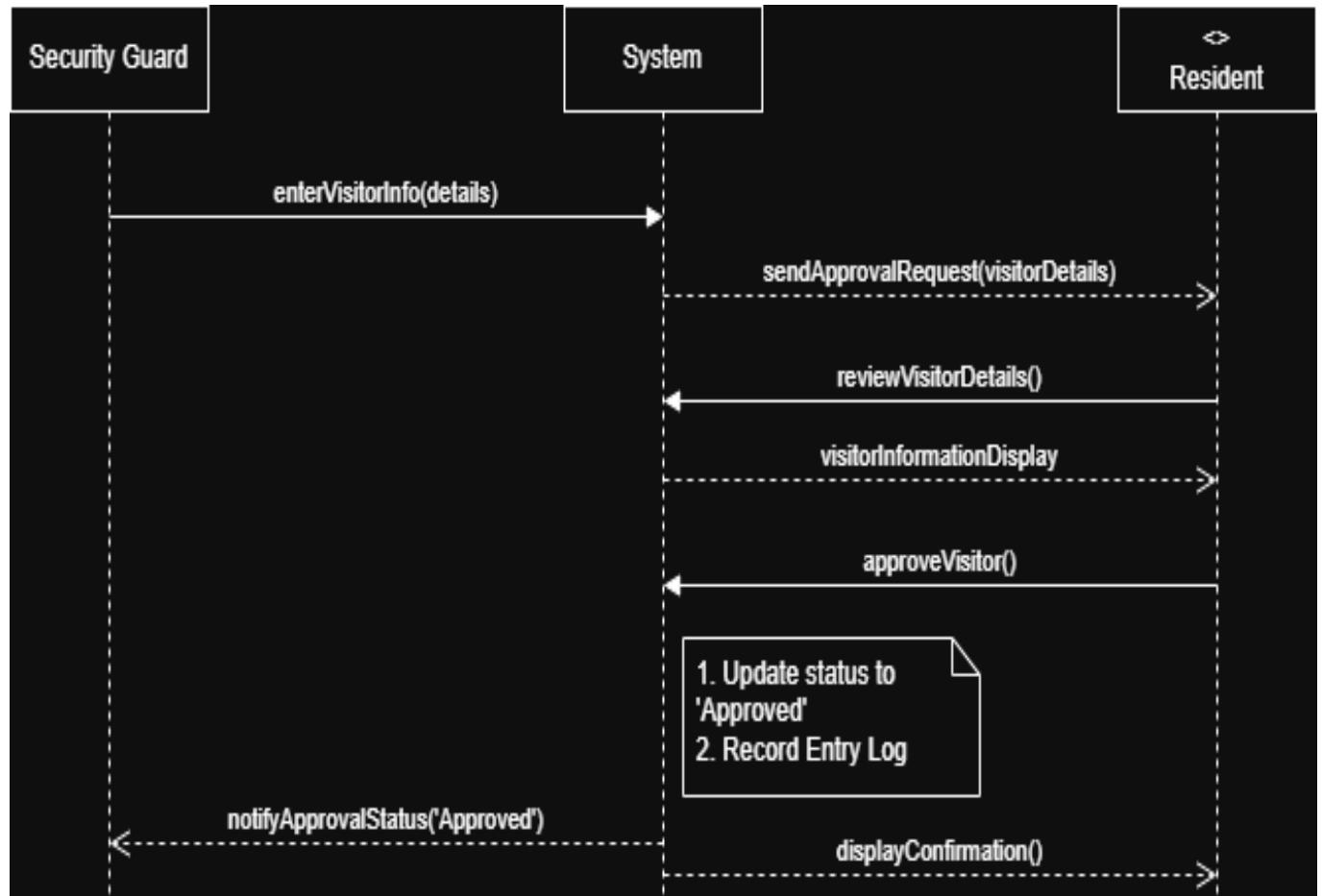
Use Case 2: Modify approval



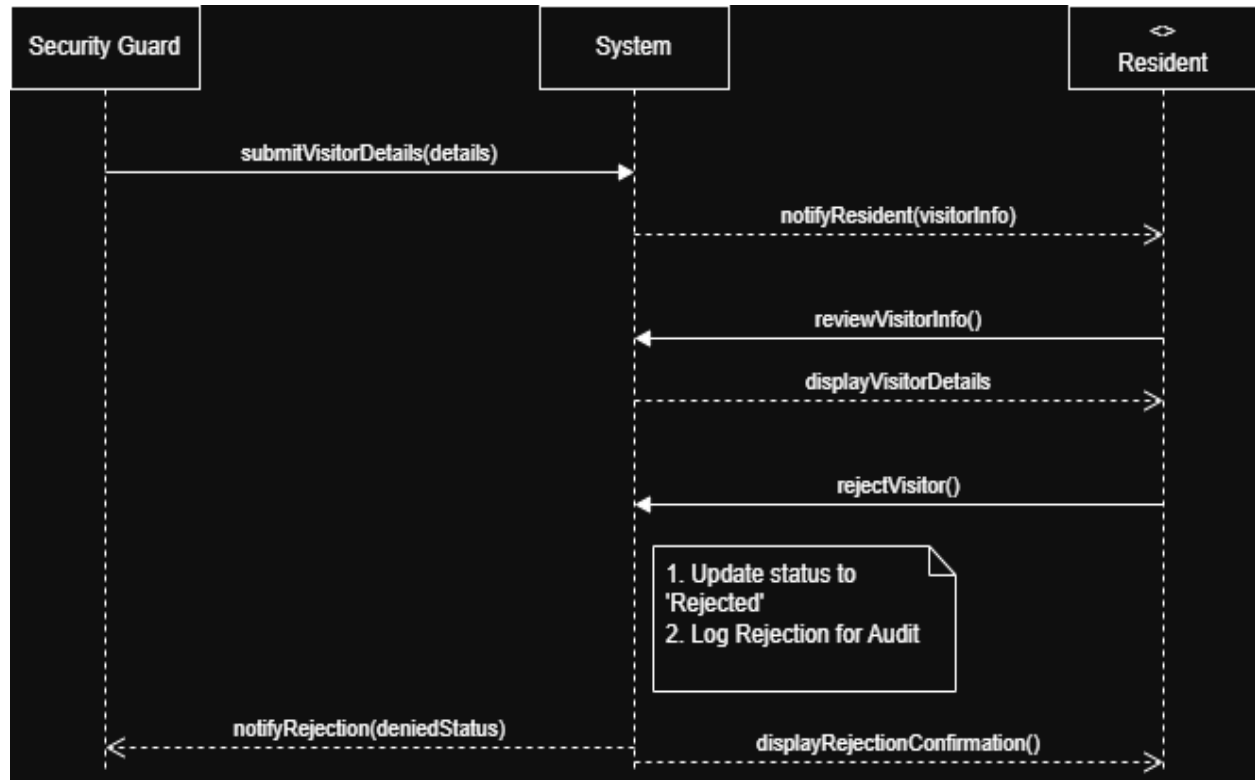
Use Case 3: Cancel Approval



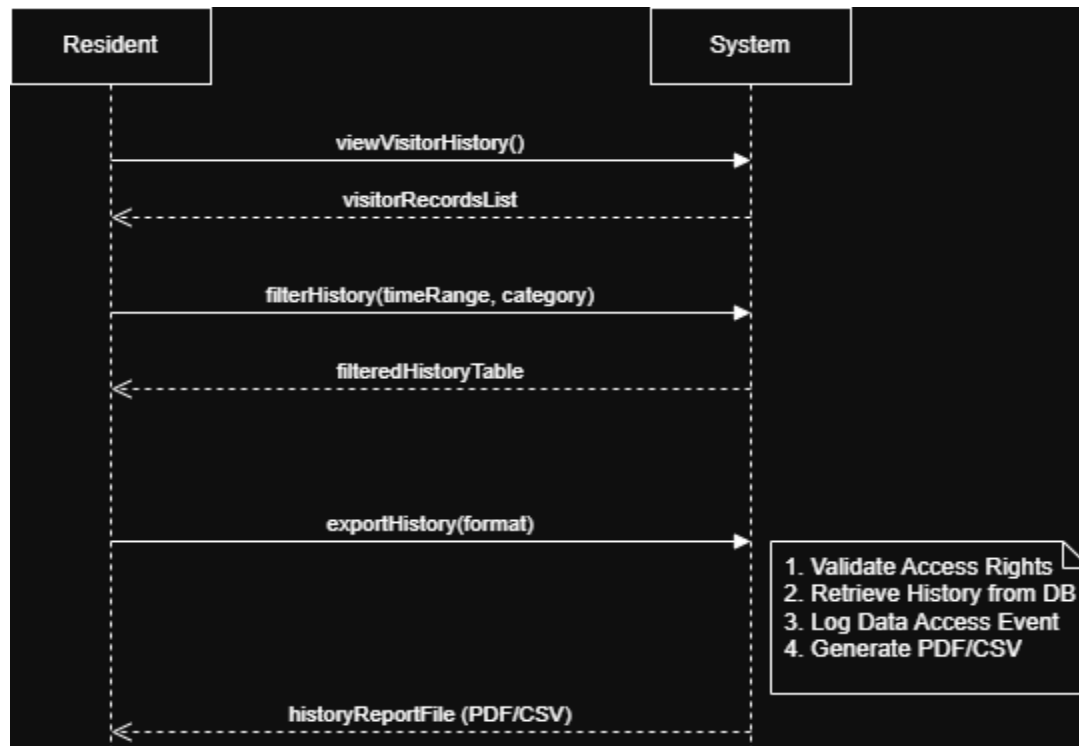
Use Case 4: Approval Visitor On Arrival



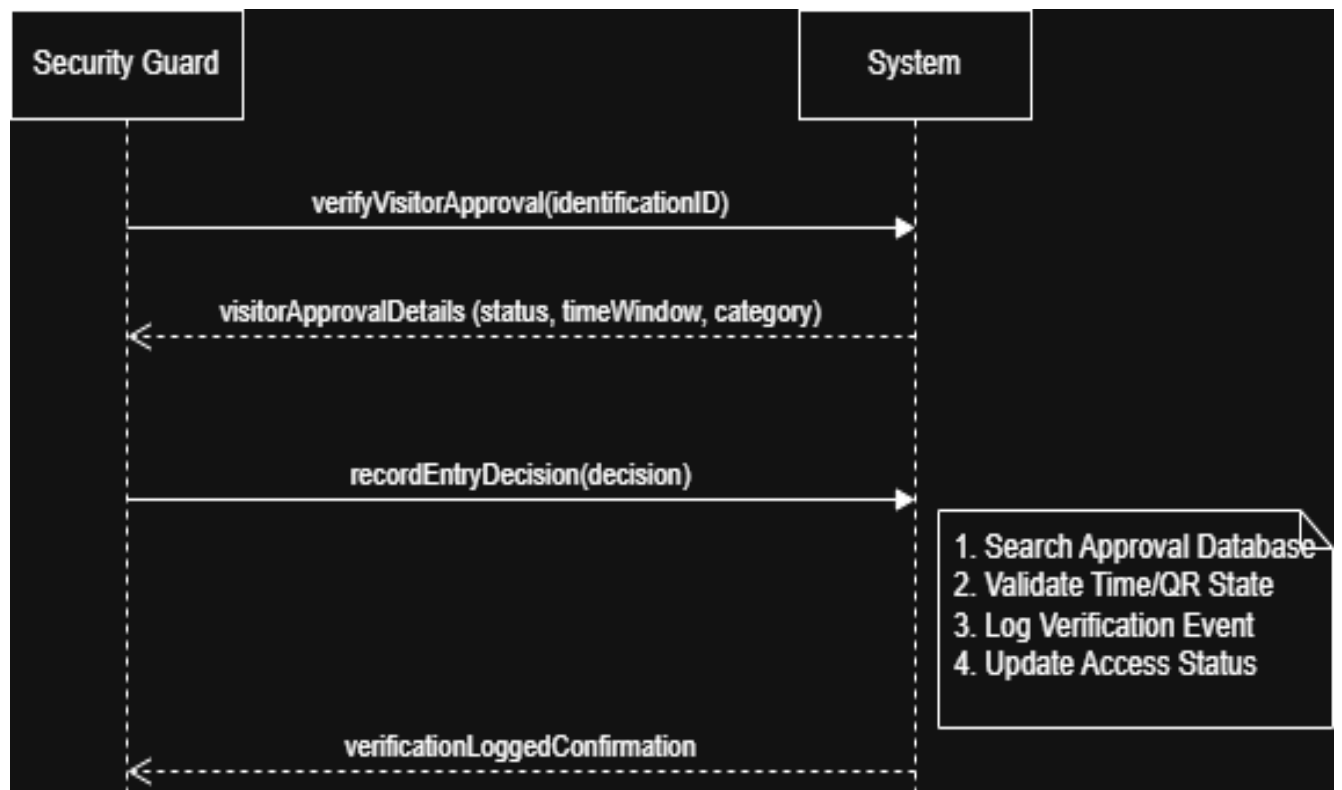
Use Case 5: Reject Visitor On Arrival



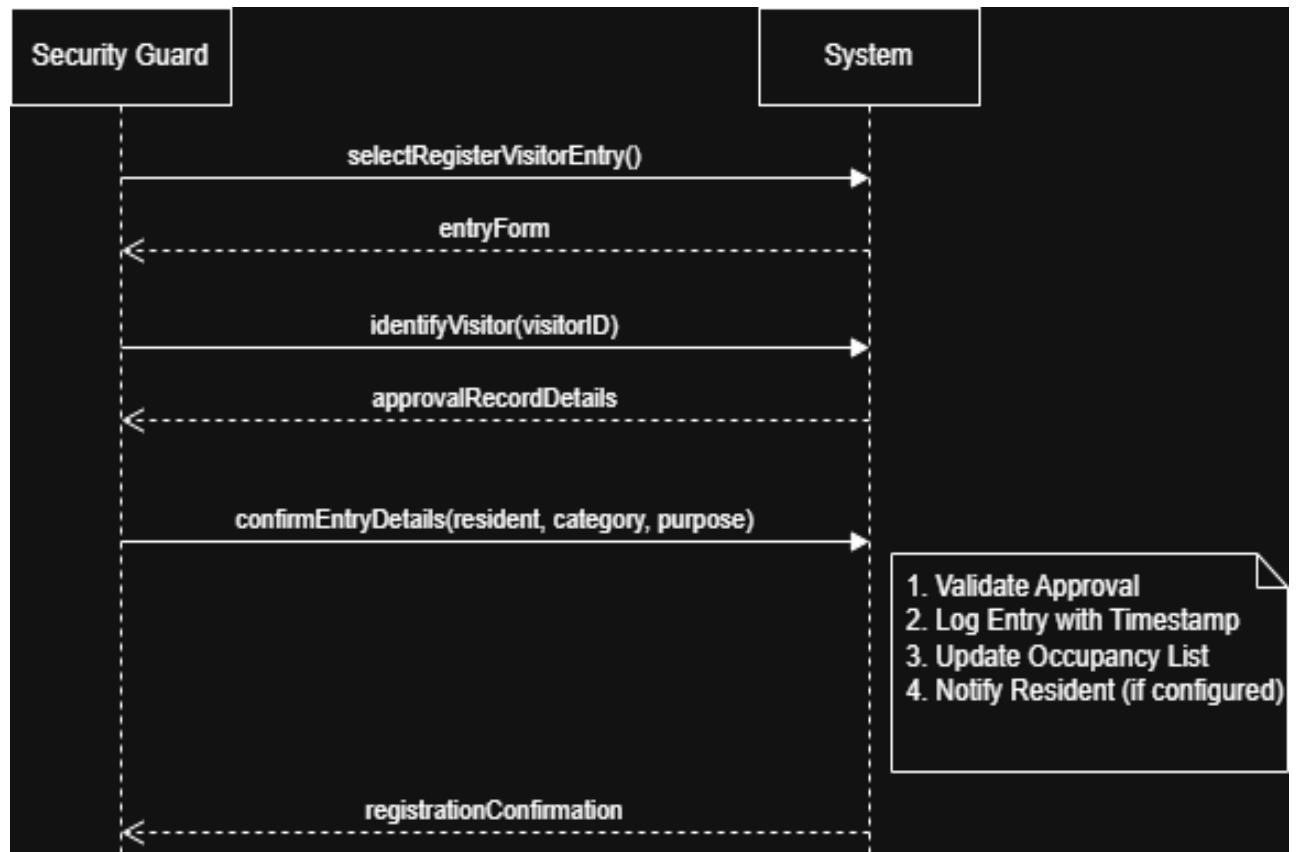
Use Case 6: View Visitor History



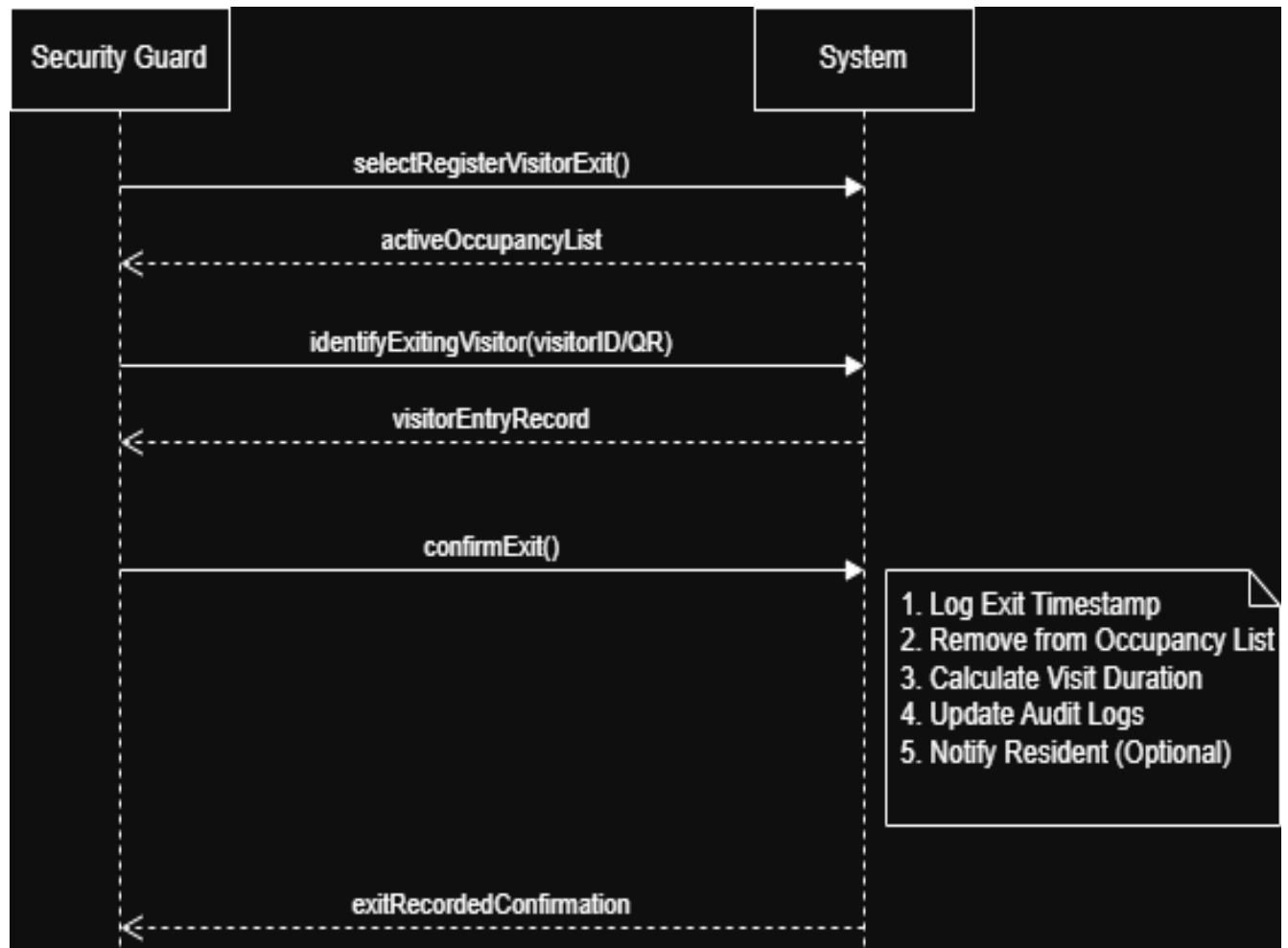
Use Case 7: Verify Visitor Approval Status



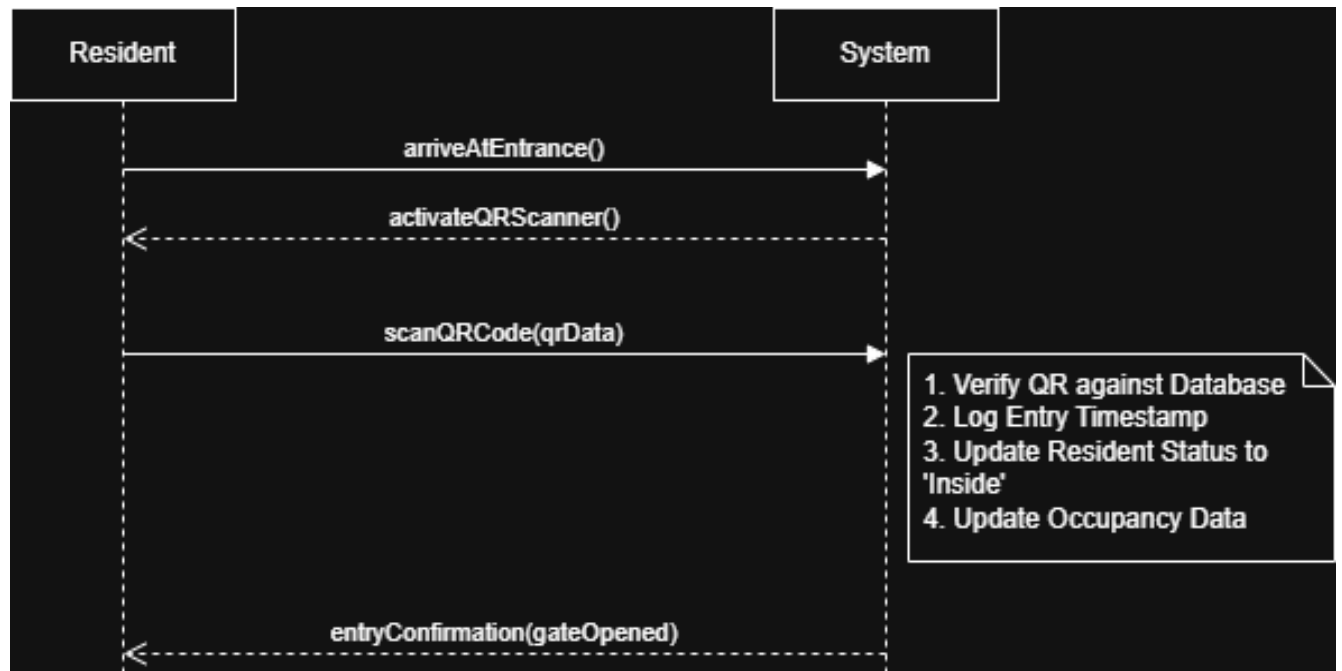
Use Case 8: Register Visitor Entry



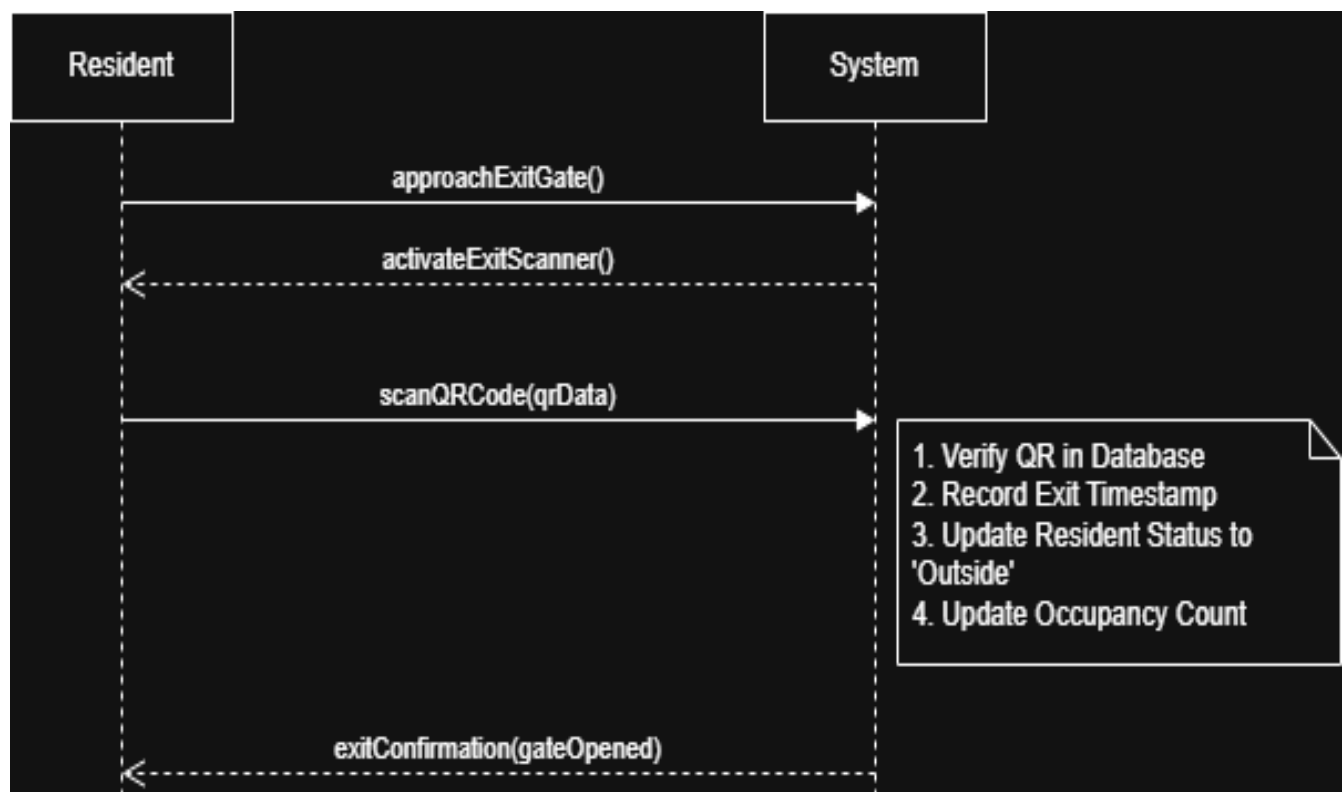
Use Case 9: Register Visitor Exit



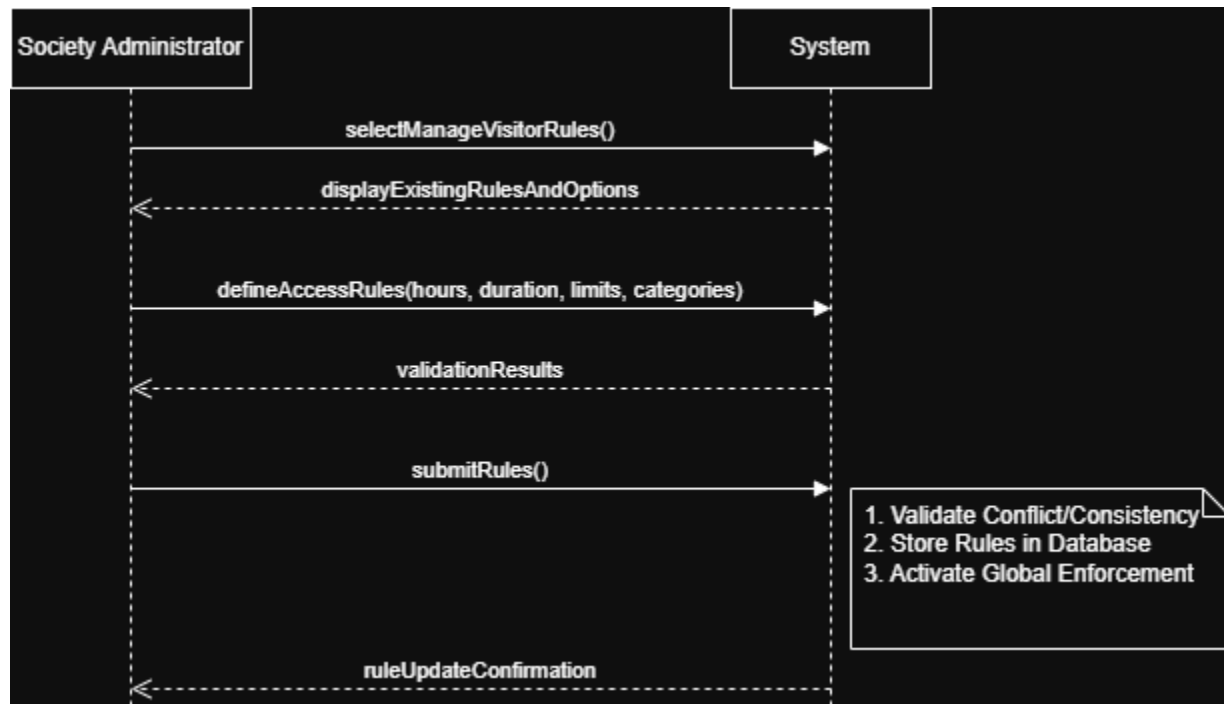
Use Case 10: Register Resident Entry



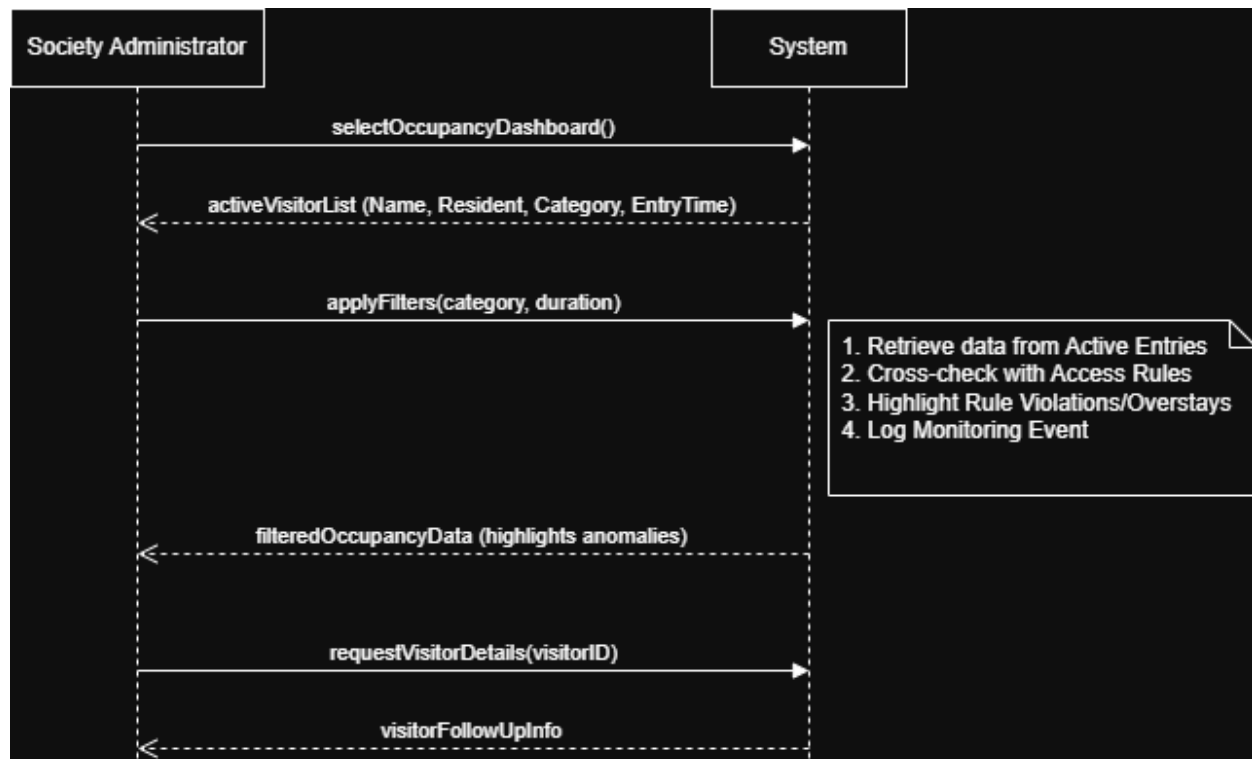
Use Case 11: Register Resident Exit



Use Case 12: Define Visitor Access Rule



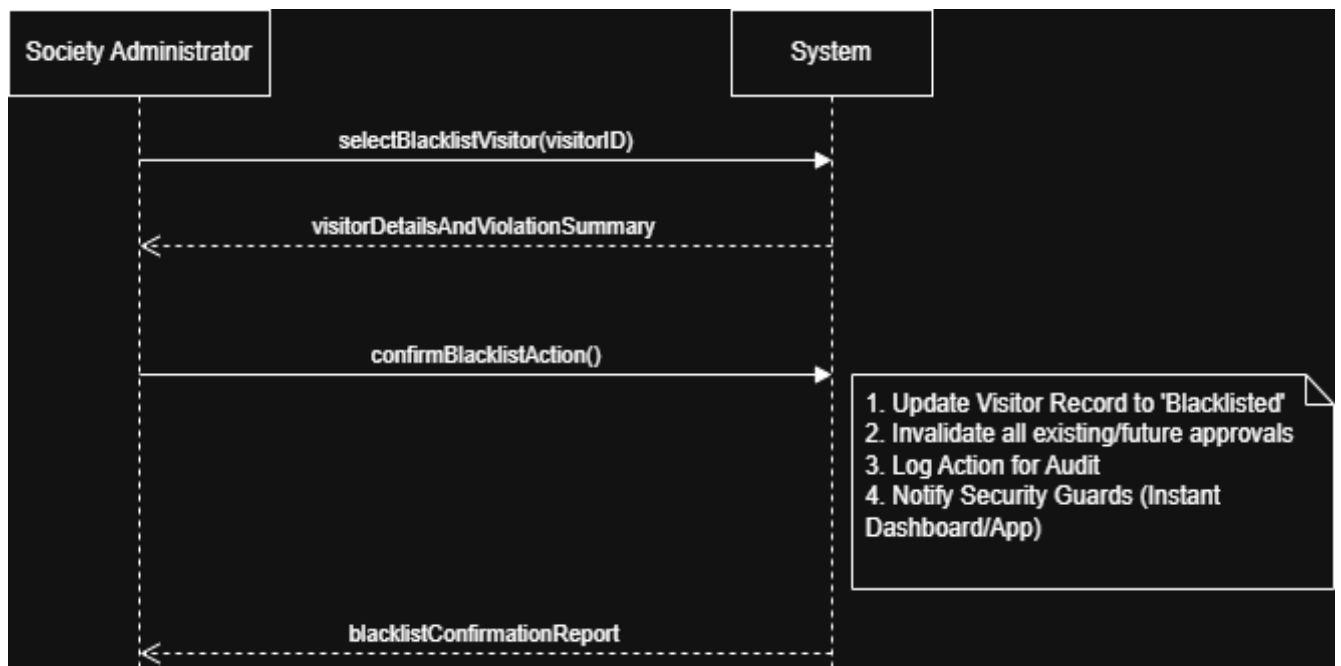
Use Case 13: Monitor Real Time Occupancy



UseCase 14: Review Violation and Take Action



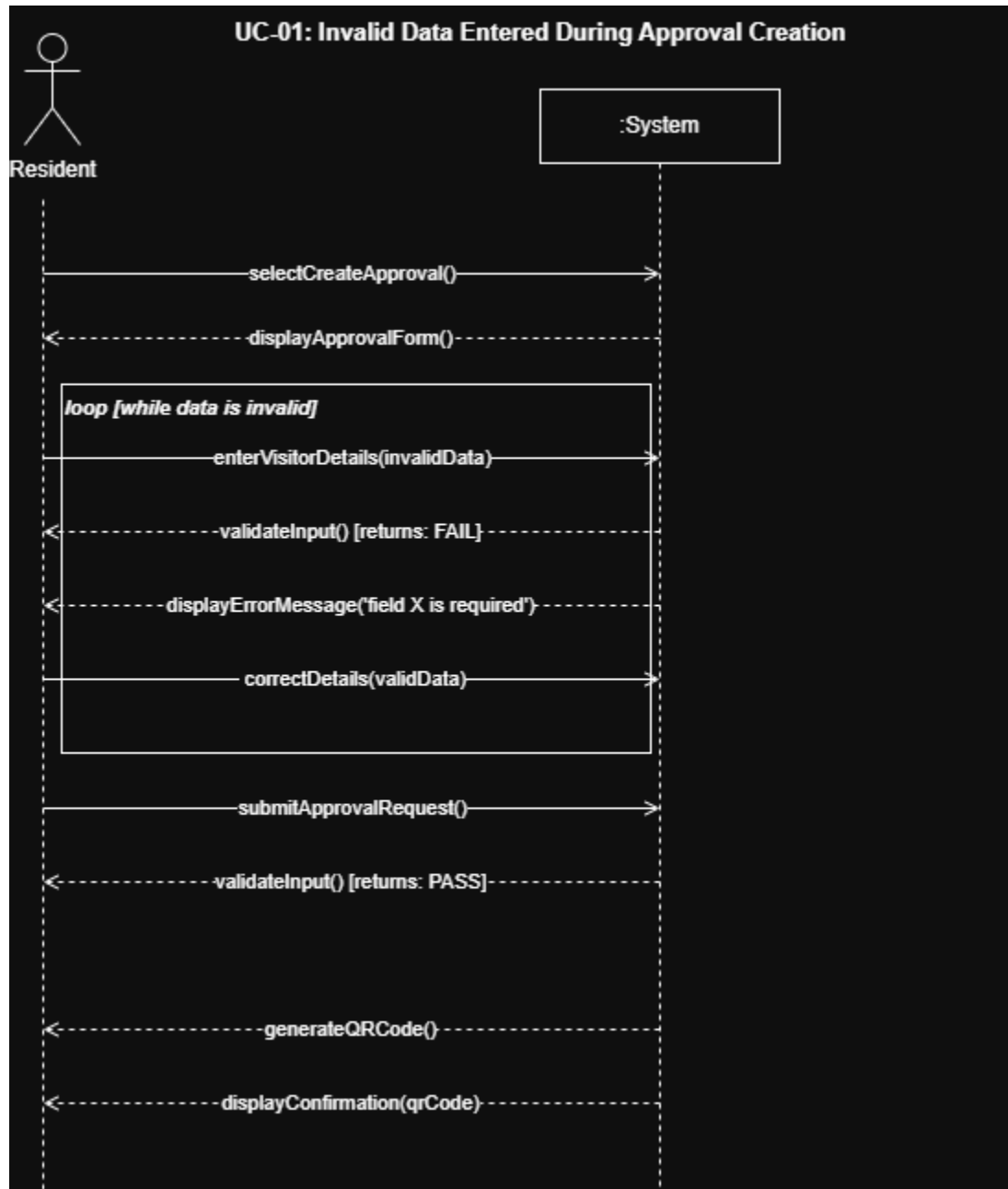
Use Case 15: BlackList Visitor



Alternative Scenarios:

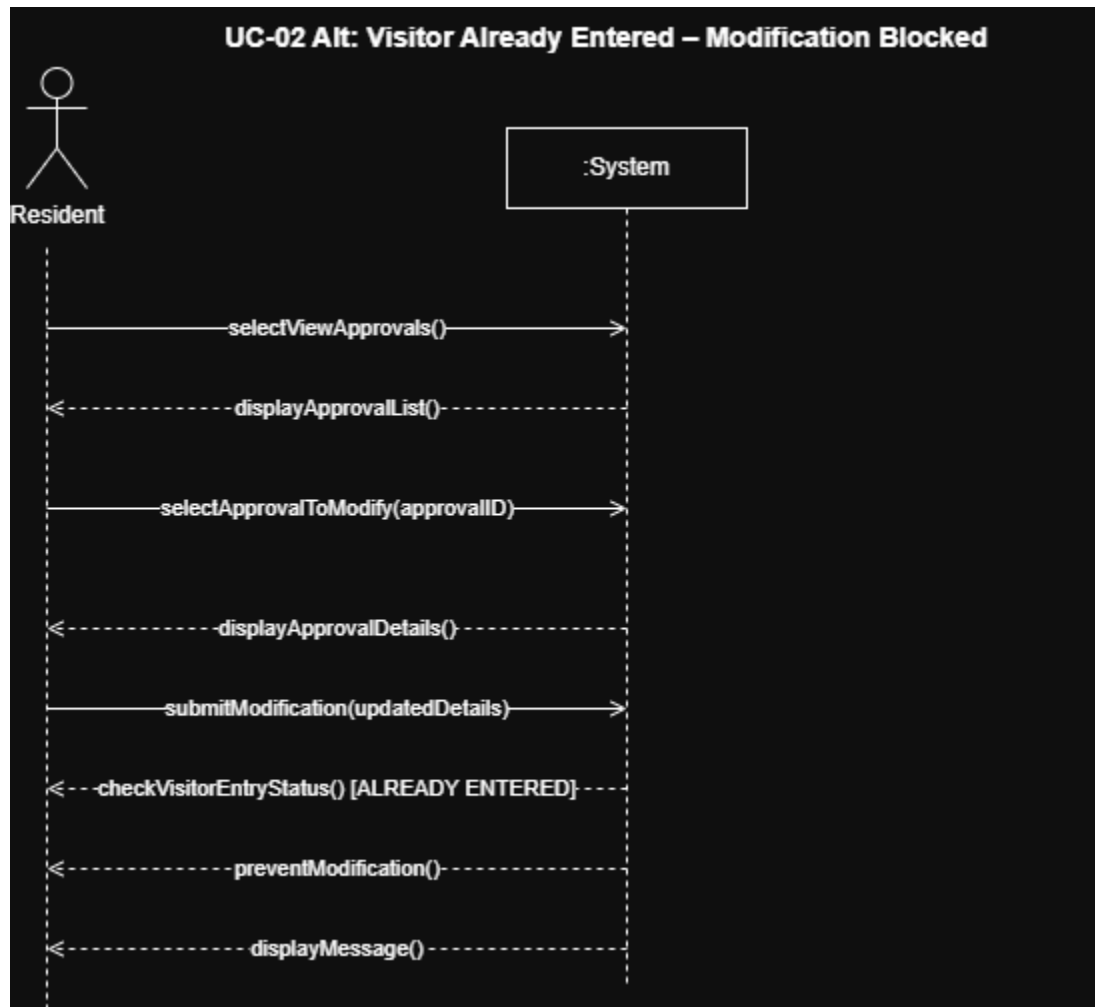
Use Case 1: Create approval

Extension : Create approval with Invalid Data



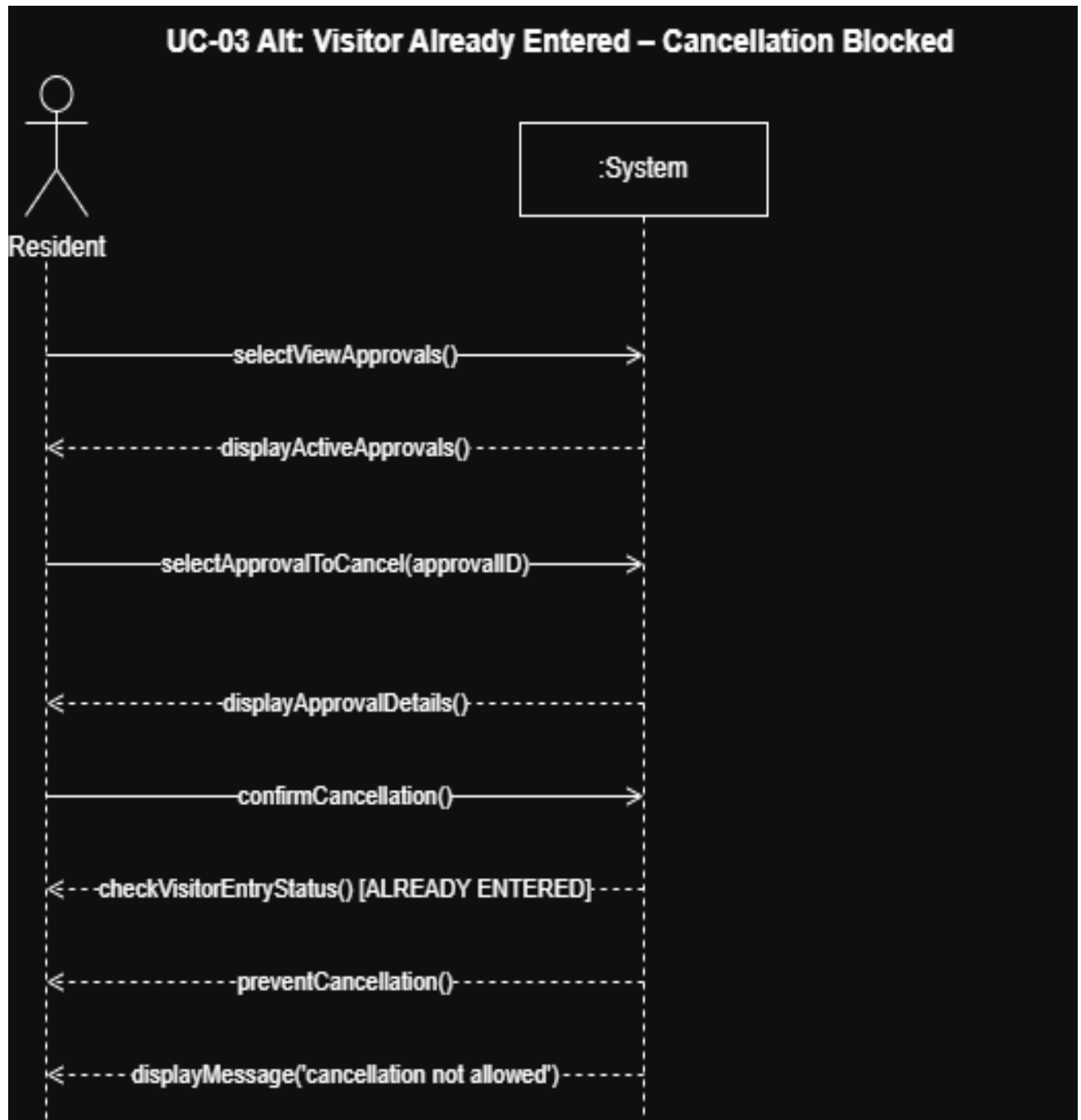
Use Case 2: Modify Approval

Extension : Already Entered Visitor



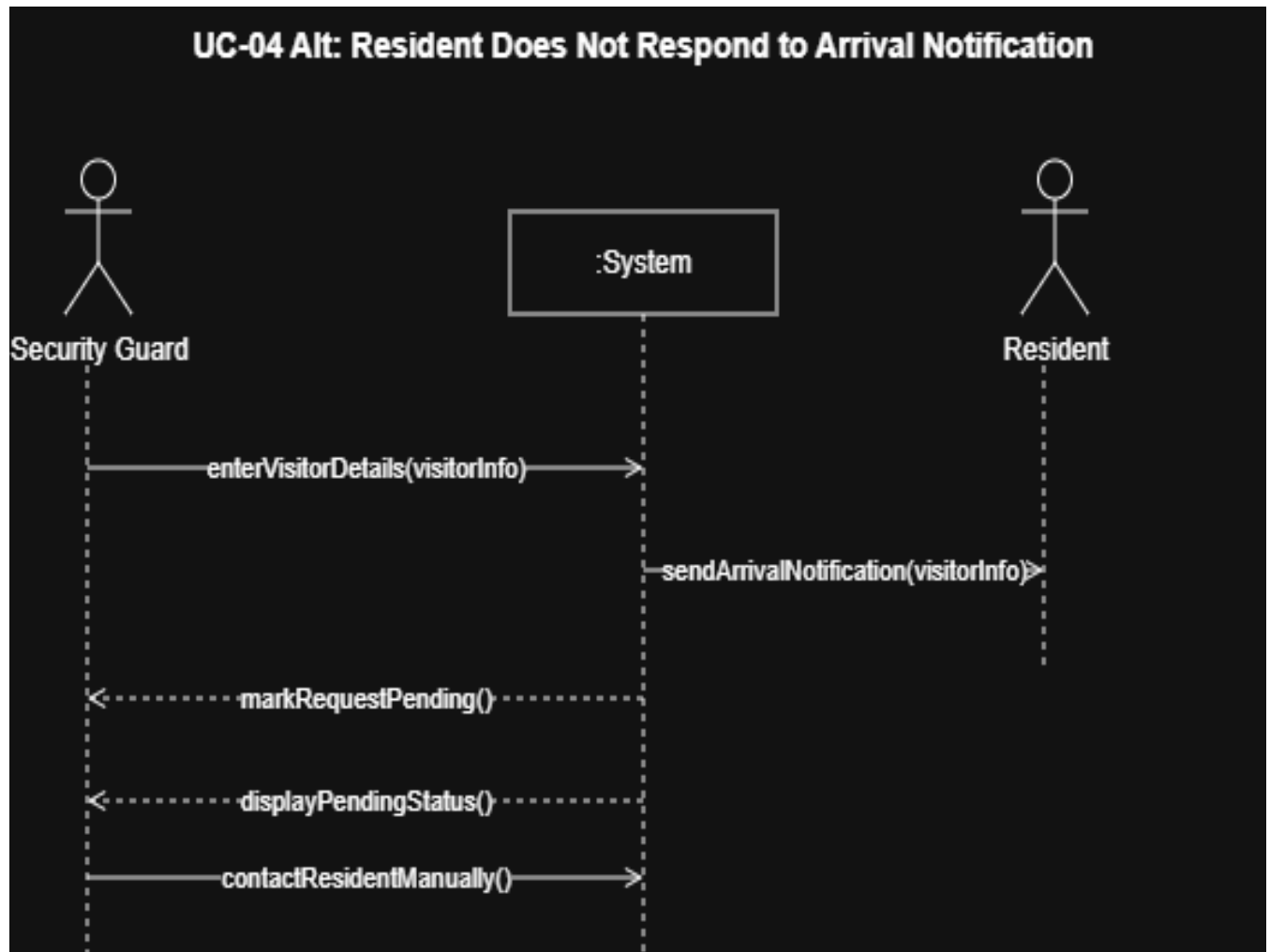
Use Case 3: Cancel Approval

Extension: Visitor Already Entered



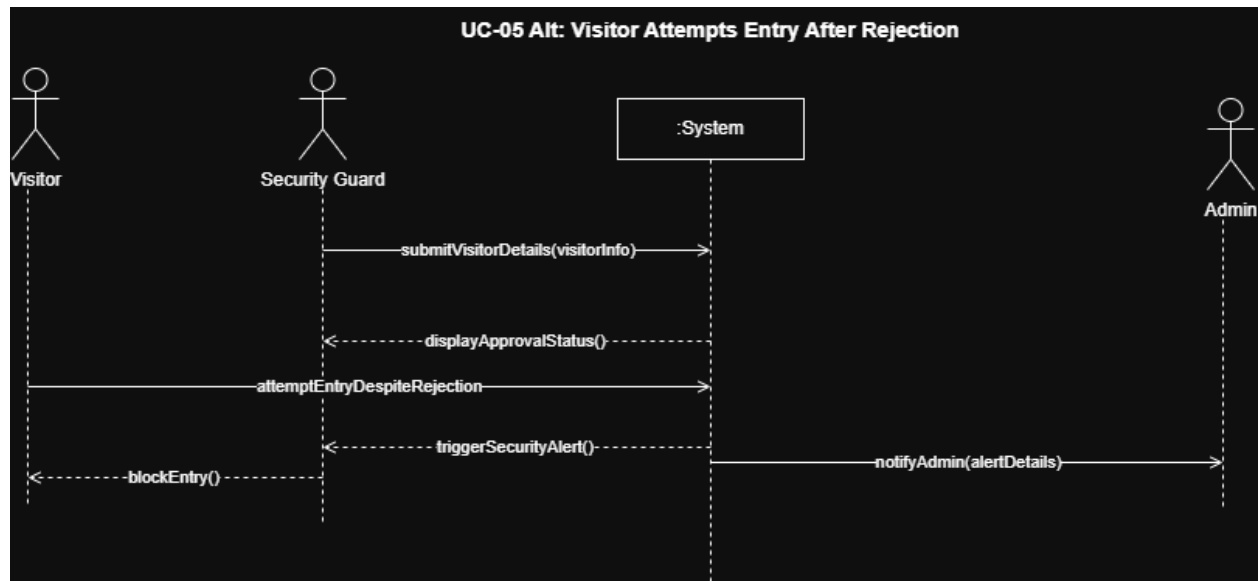
Use Case 4: Approval Visitor On Arrival

Extension : Resident Doesn't Respond



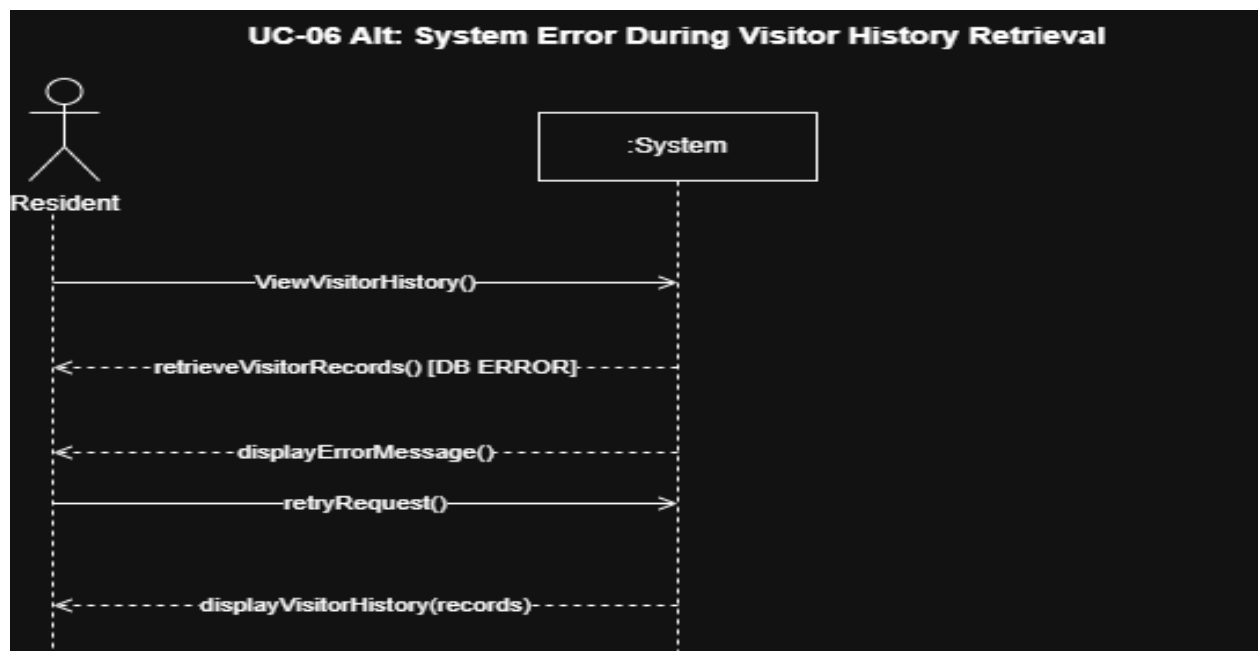
Use Case 5: Reject Visitor On Arrival

Extension : Visitor Tries to Enter Despite Rejection



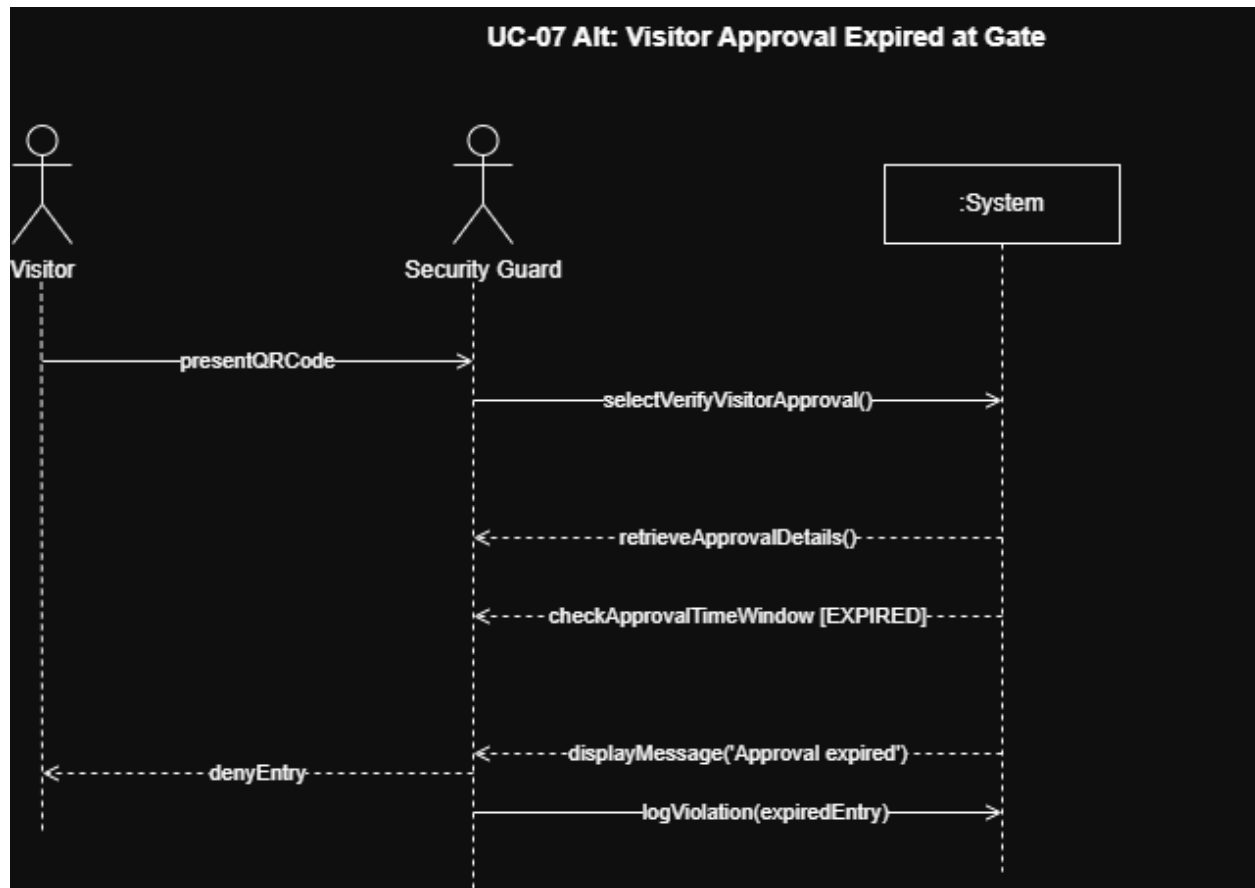
Use Case 6: View Visitor History

Extension: System Error Occur



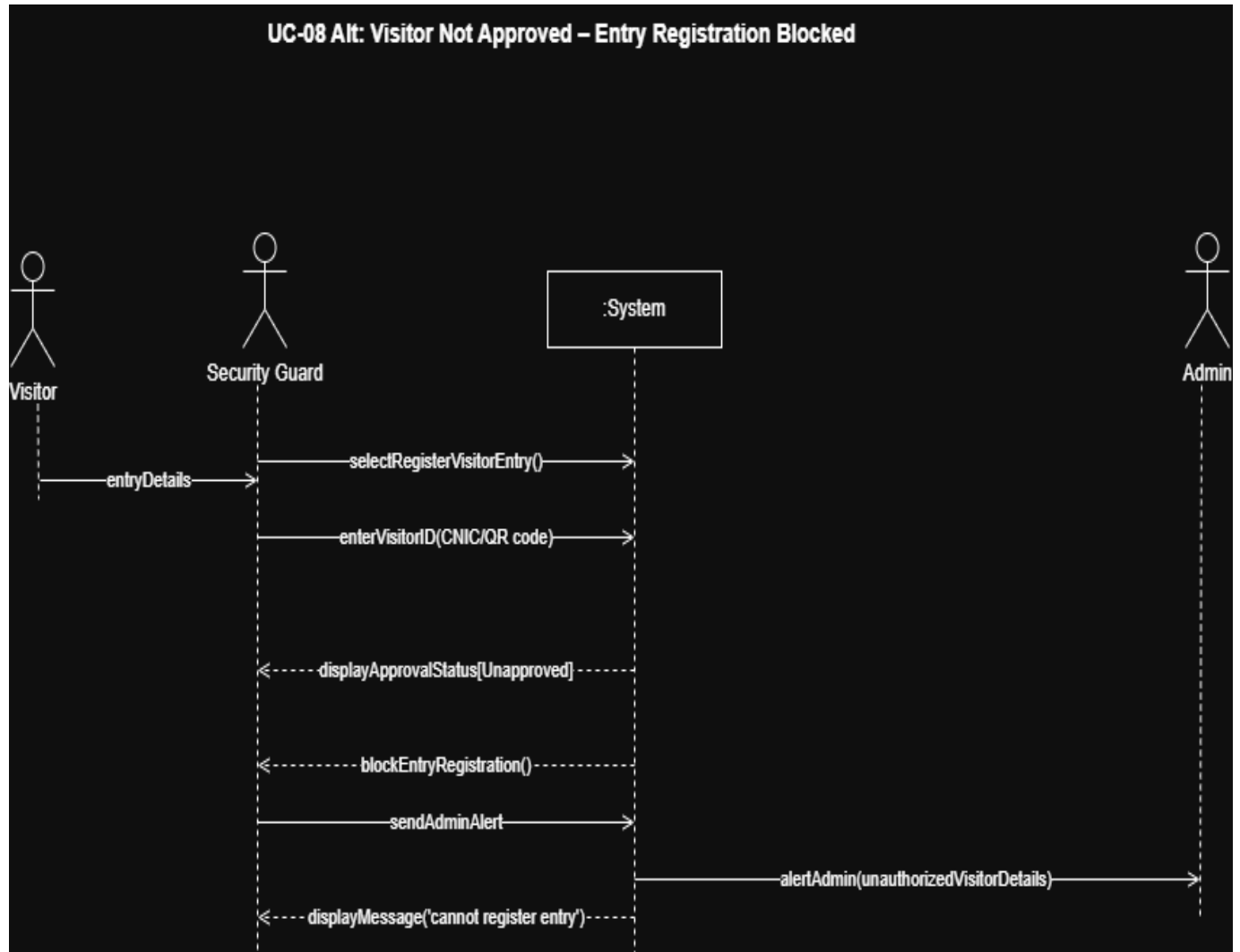
Use Case 7: Verify Visitor Approval Status

Extension: Approval Expired



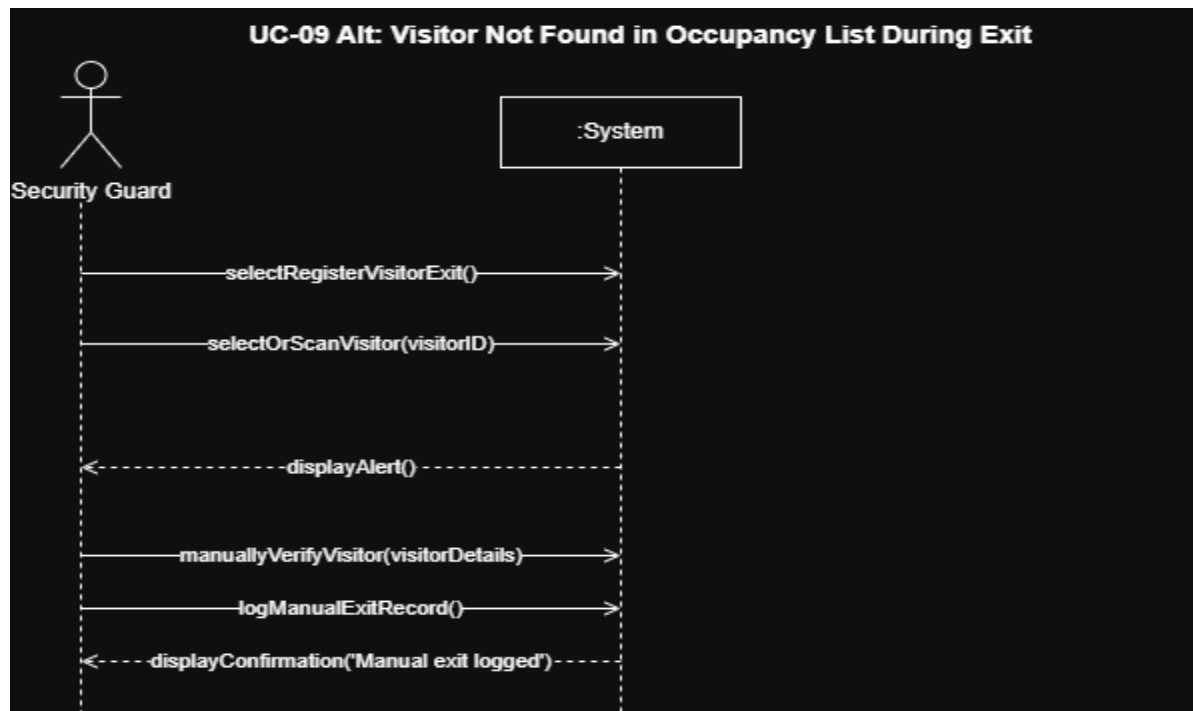
Use Case 8: Register Visitor Entry

Extension: Visitor Not Approved



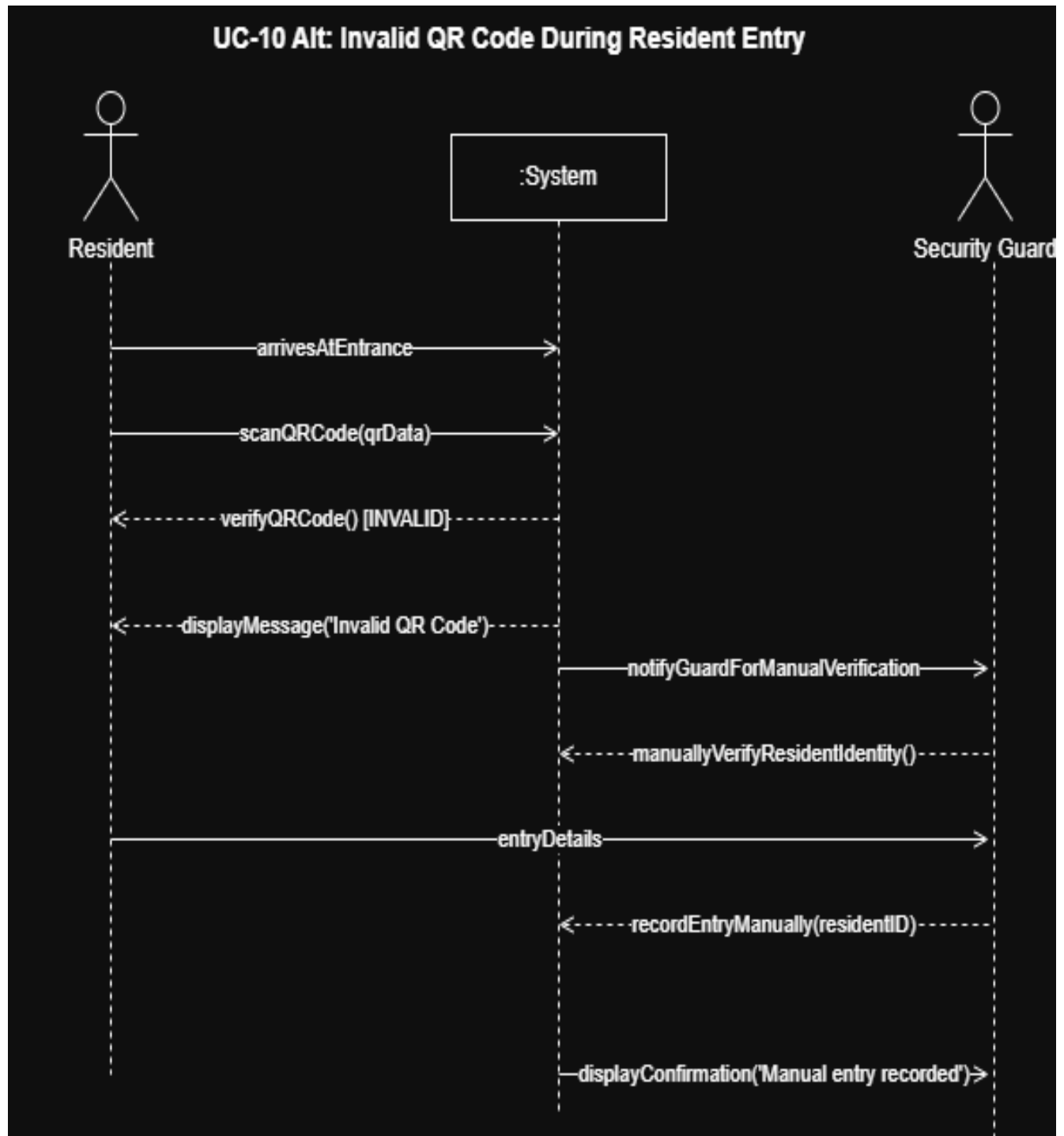
Use Case 9: Register Visitor Exit

Extension: Visitor Not Found in OccupancyList



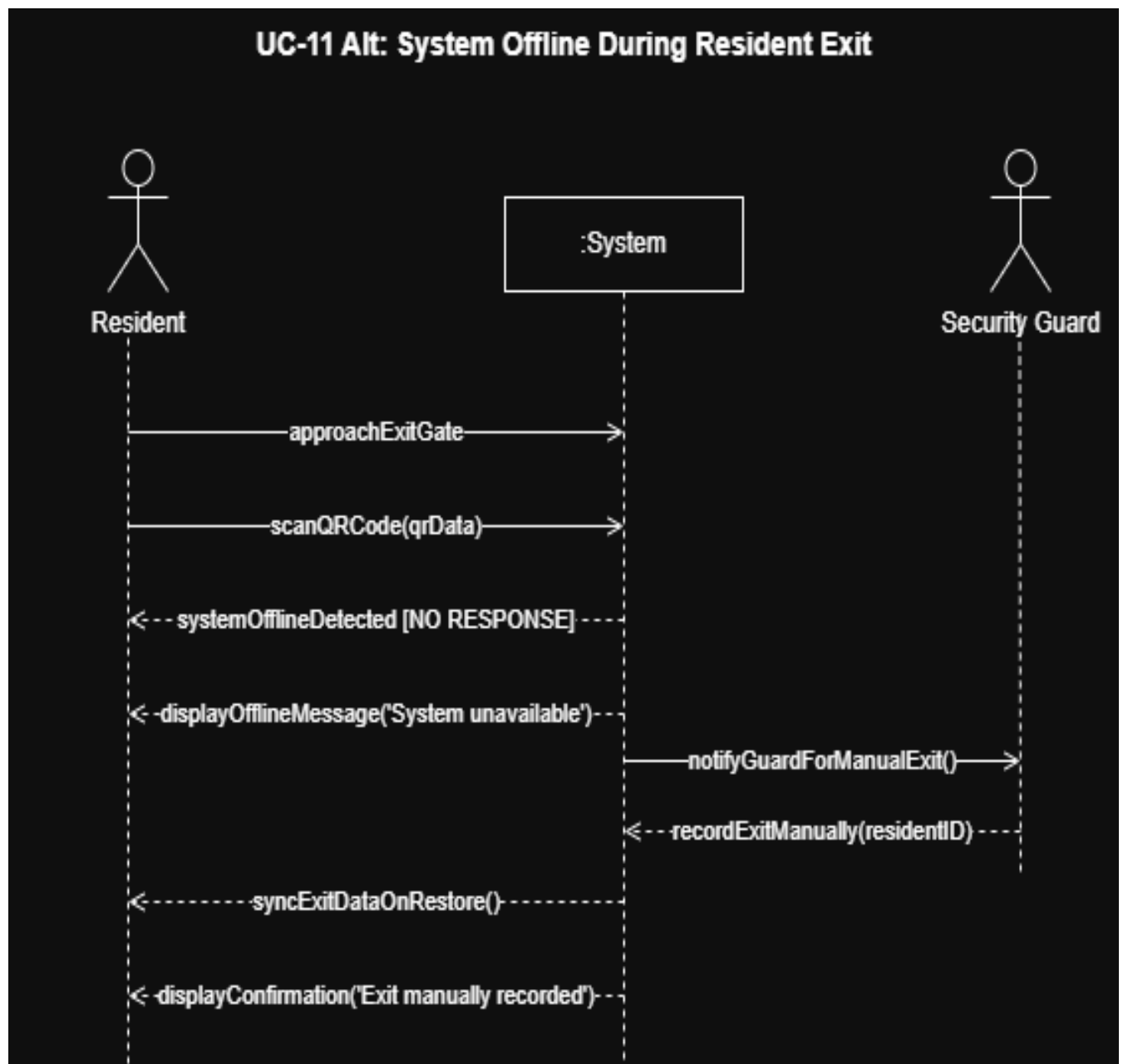
Use Case 10: Register Resident Entry

Extension: Invalid QR Code



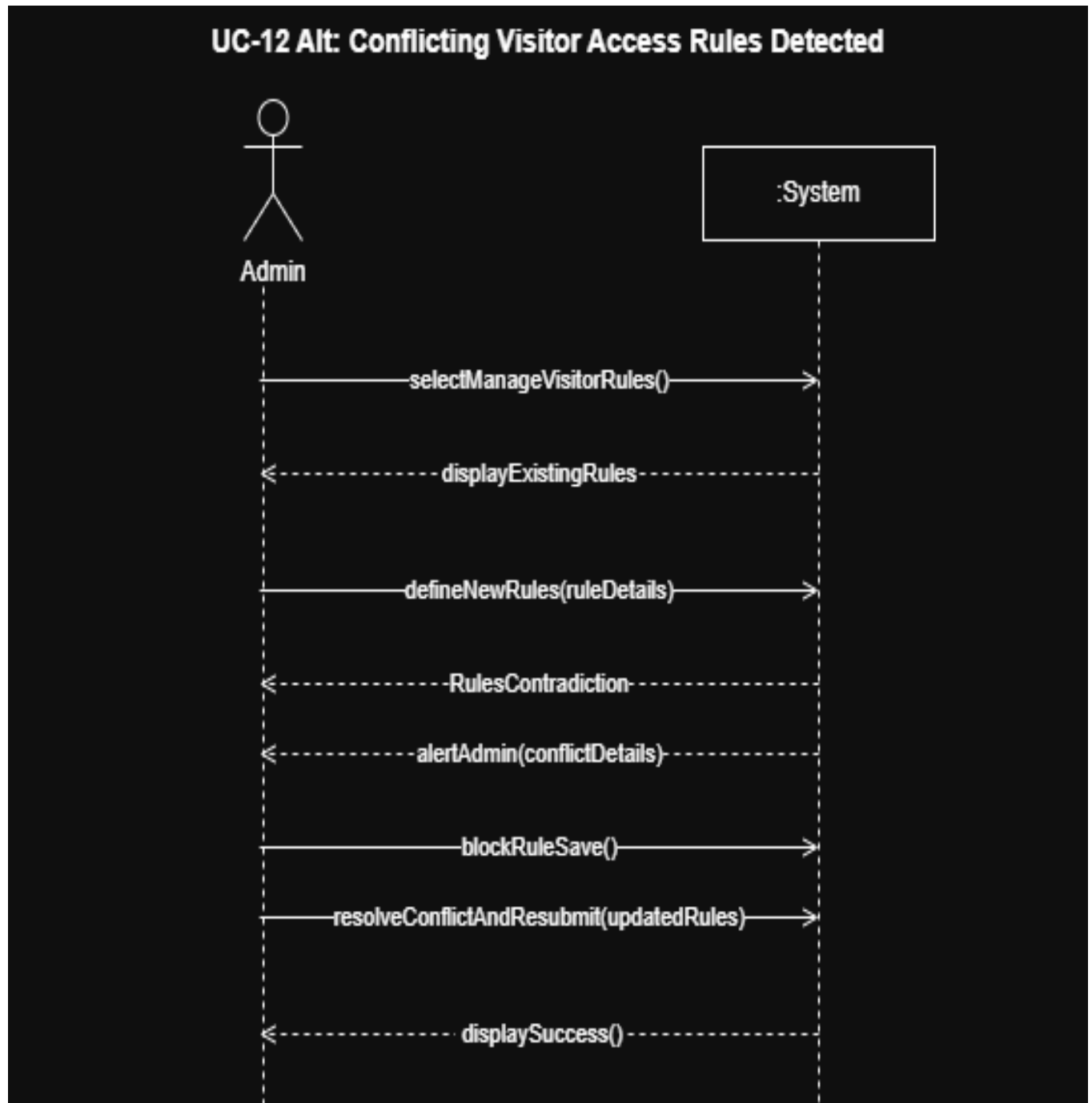
Use Case 11: Register Resident Exit

Extension: System Offline



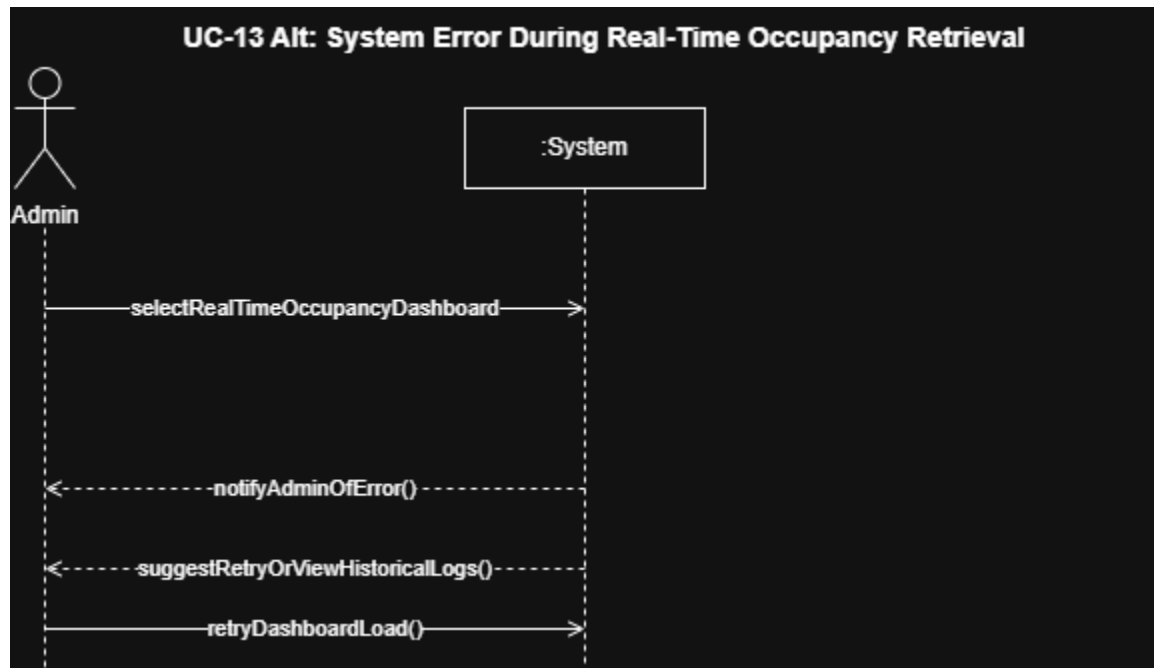
Use Case 12: Define Visitor Access Rule

Extension: Conflicting Rules Detected



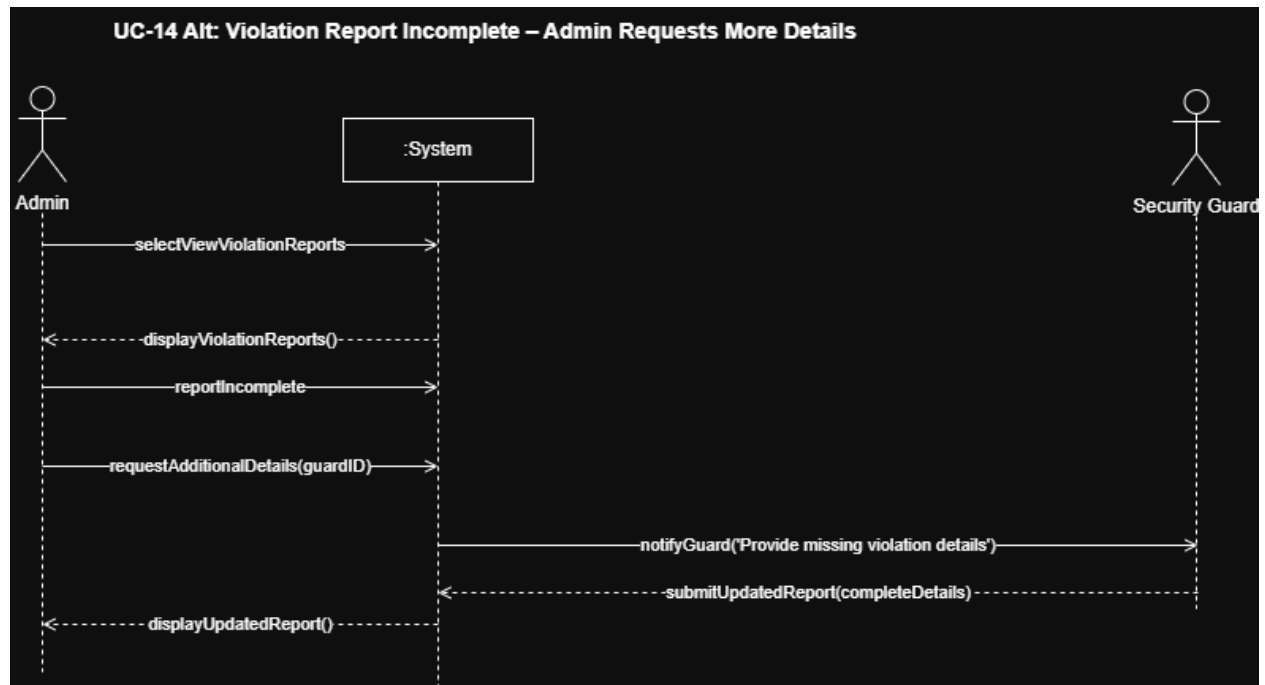
Use Case 13: Monitor Real Time Occupancy

Extension: System Error in Retrieval



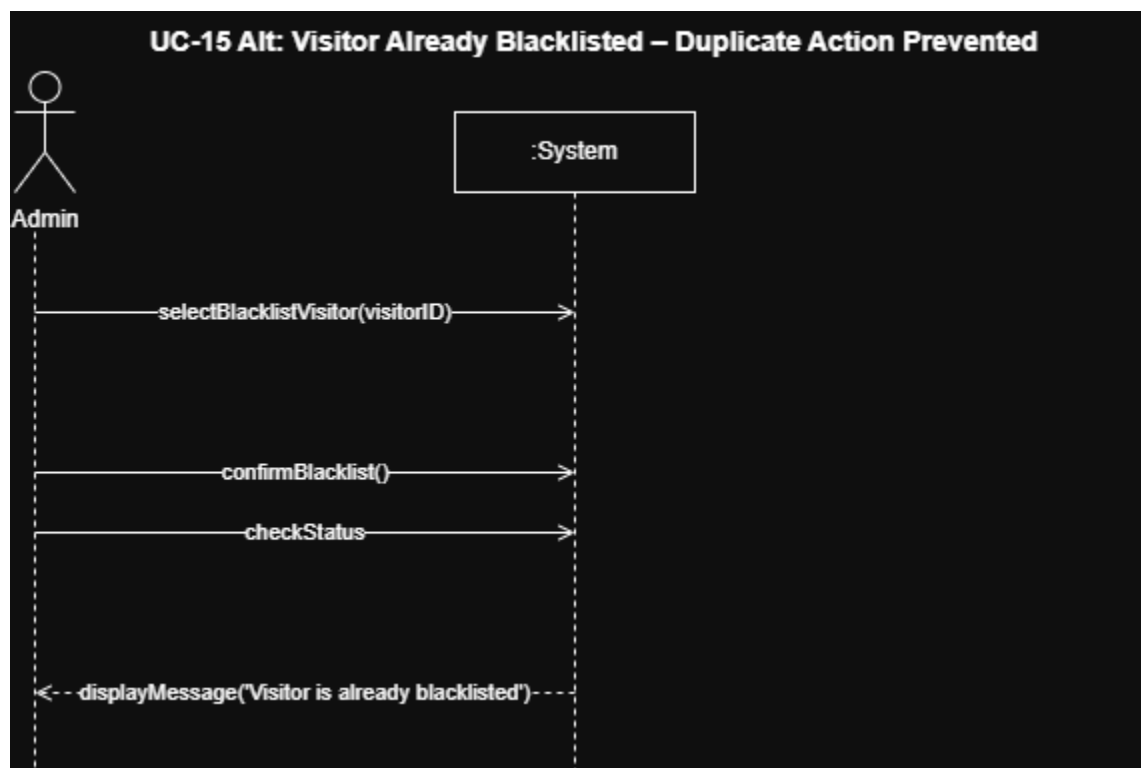
UseCase 14: Review Violation and Take Action

Extension: Violation Report Incomplete



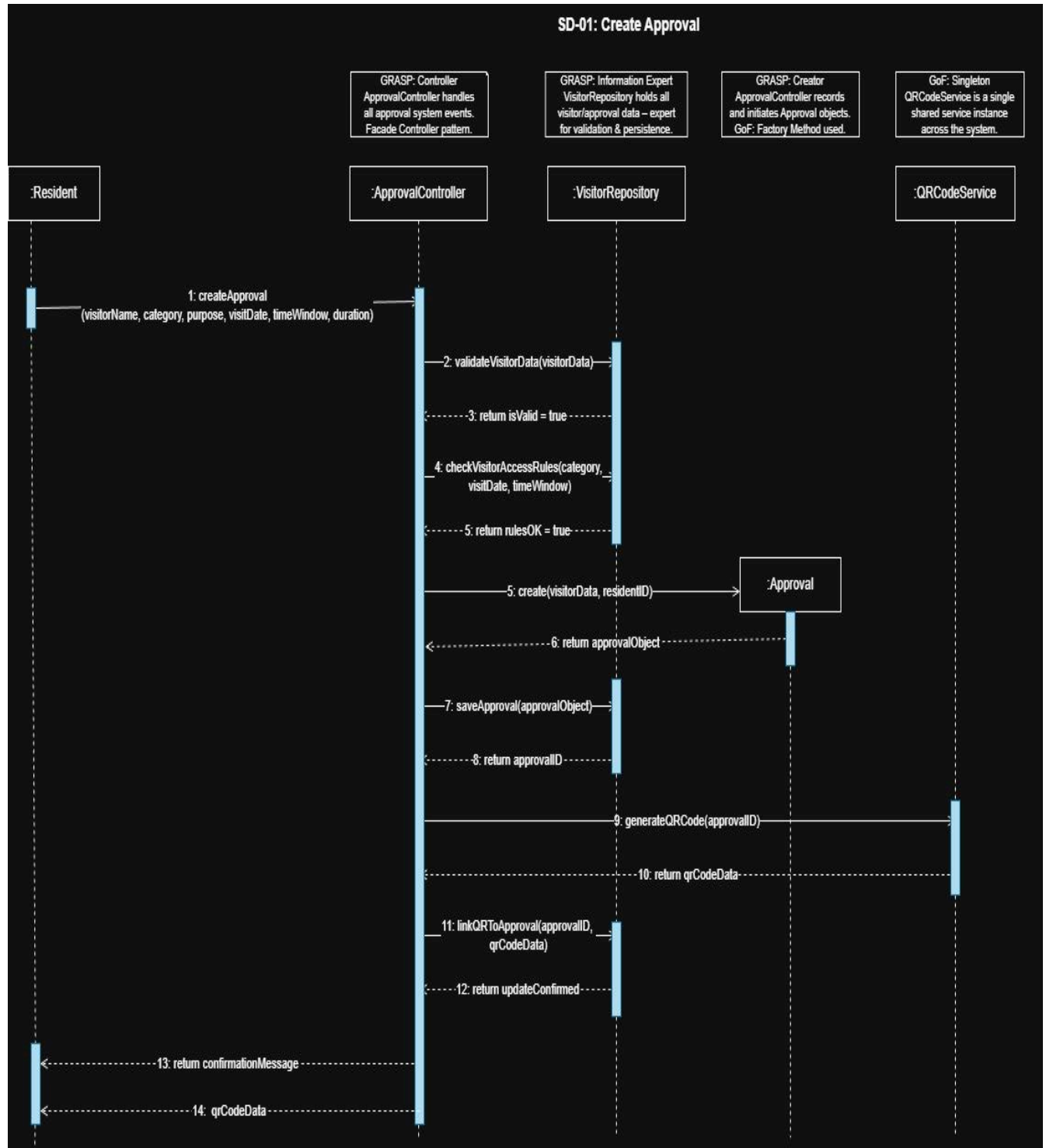
Use Case 15: BlackList Visitor

Extension: Visitor Already Blacklisted

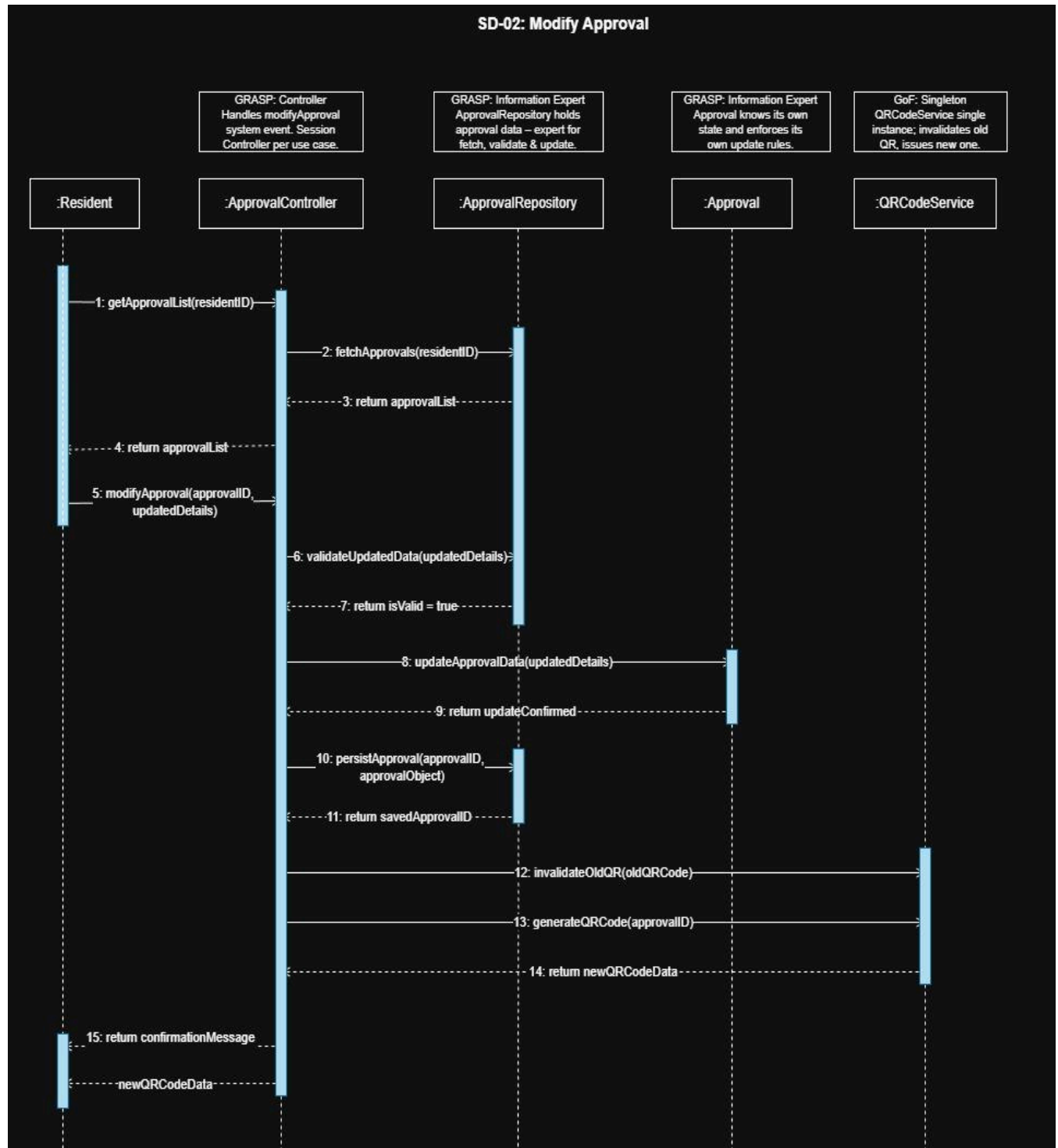


Sequence Diagrams:

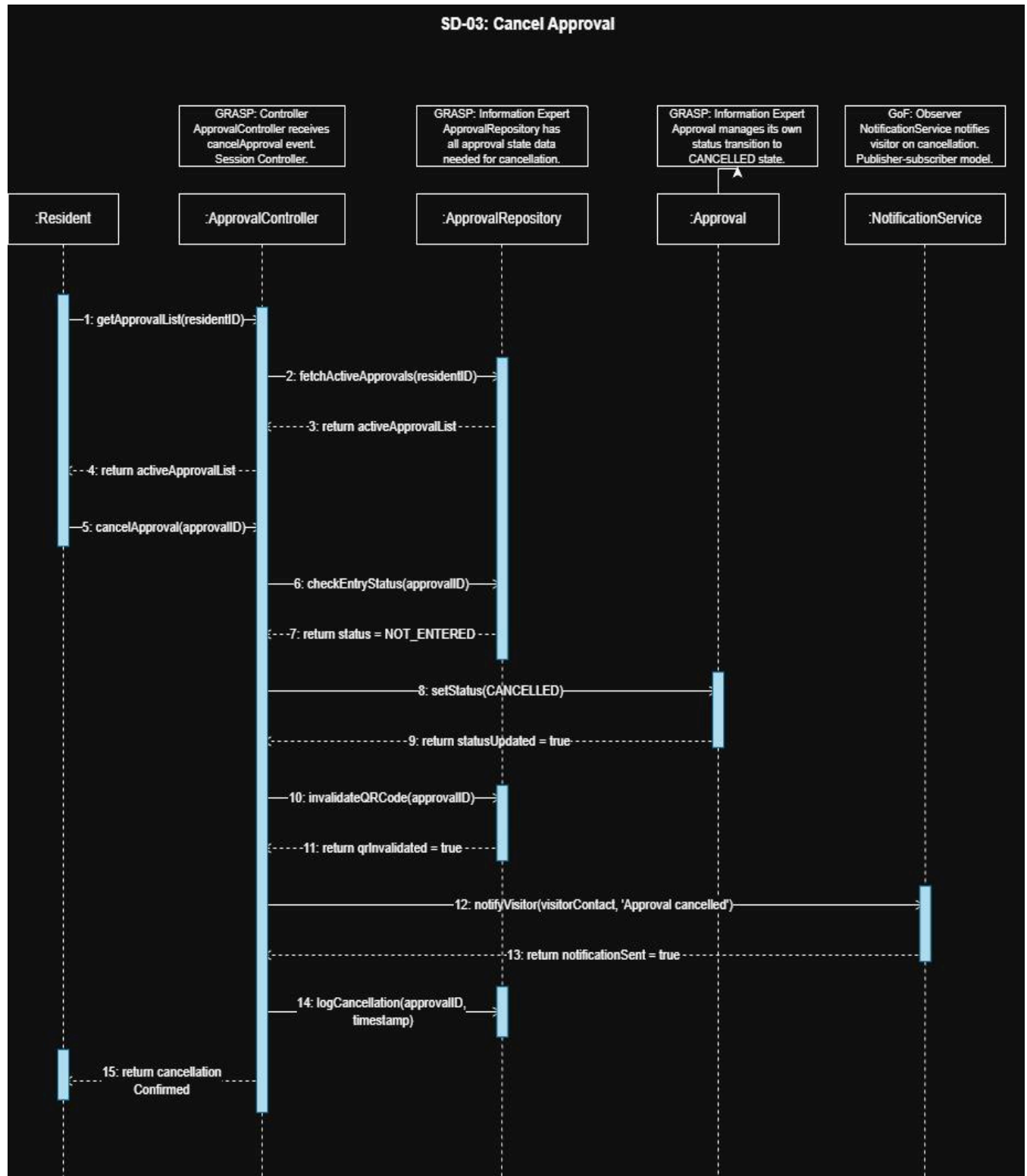
Use Case 1: Create approval



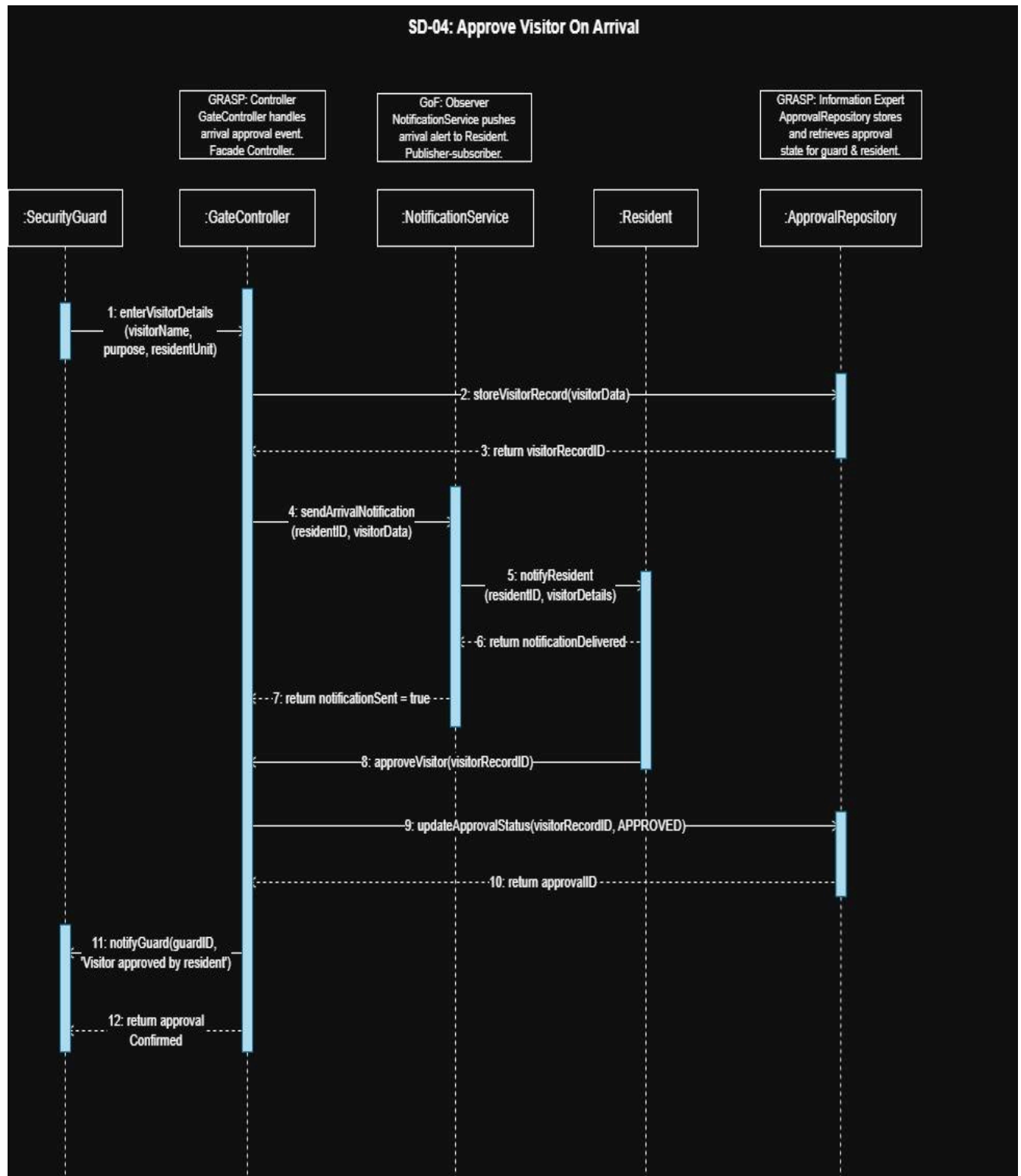
Use Case 2: Modify approval



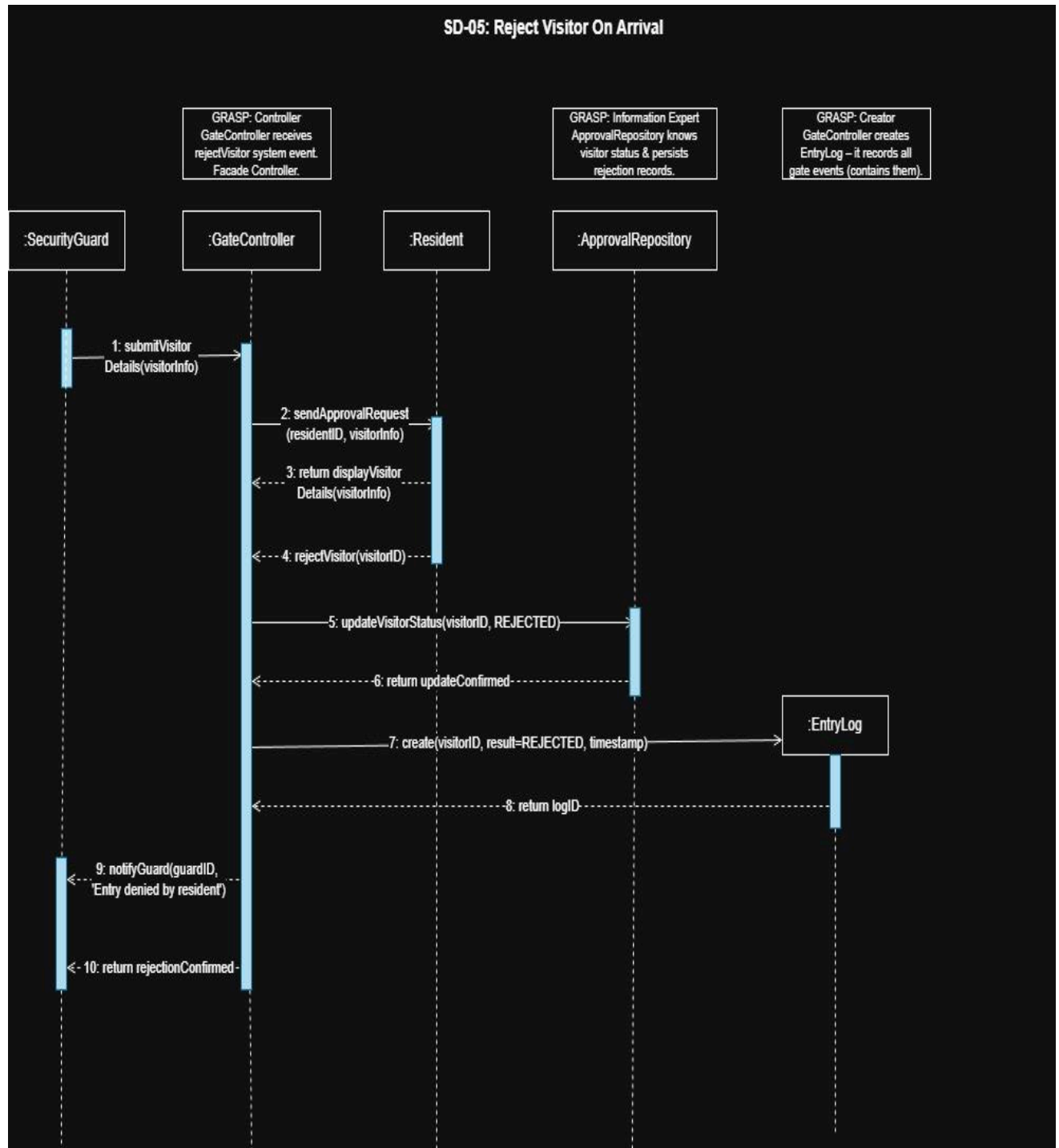
Use Case 3: Cancel Approval



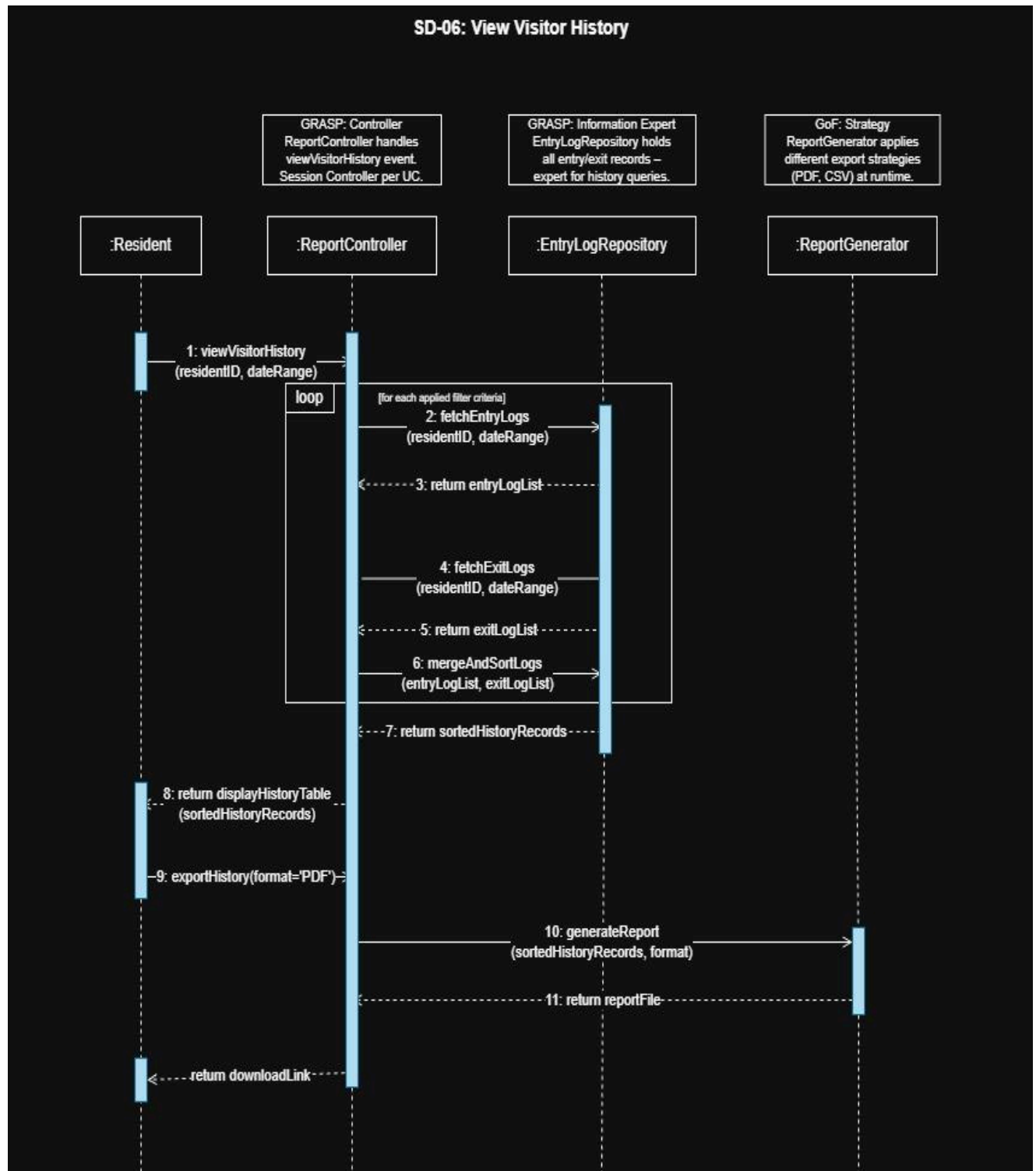
Use Case 4: Approval Visitor On Arrival



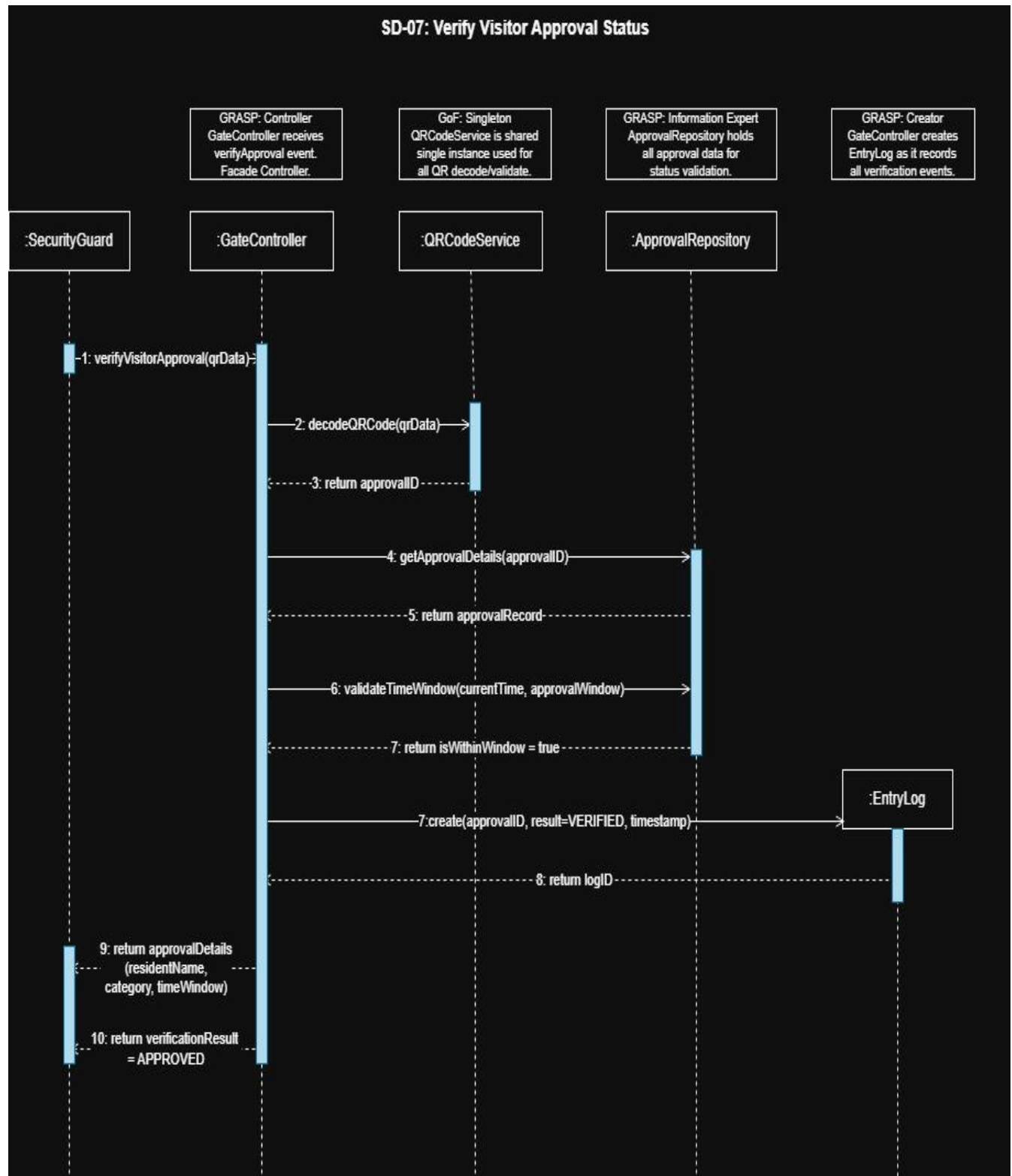
Use Case 5: Approval Visitor On Arrival



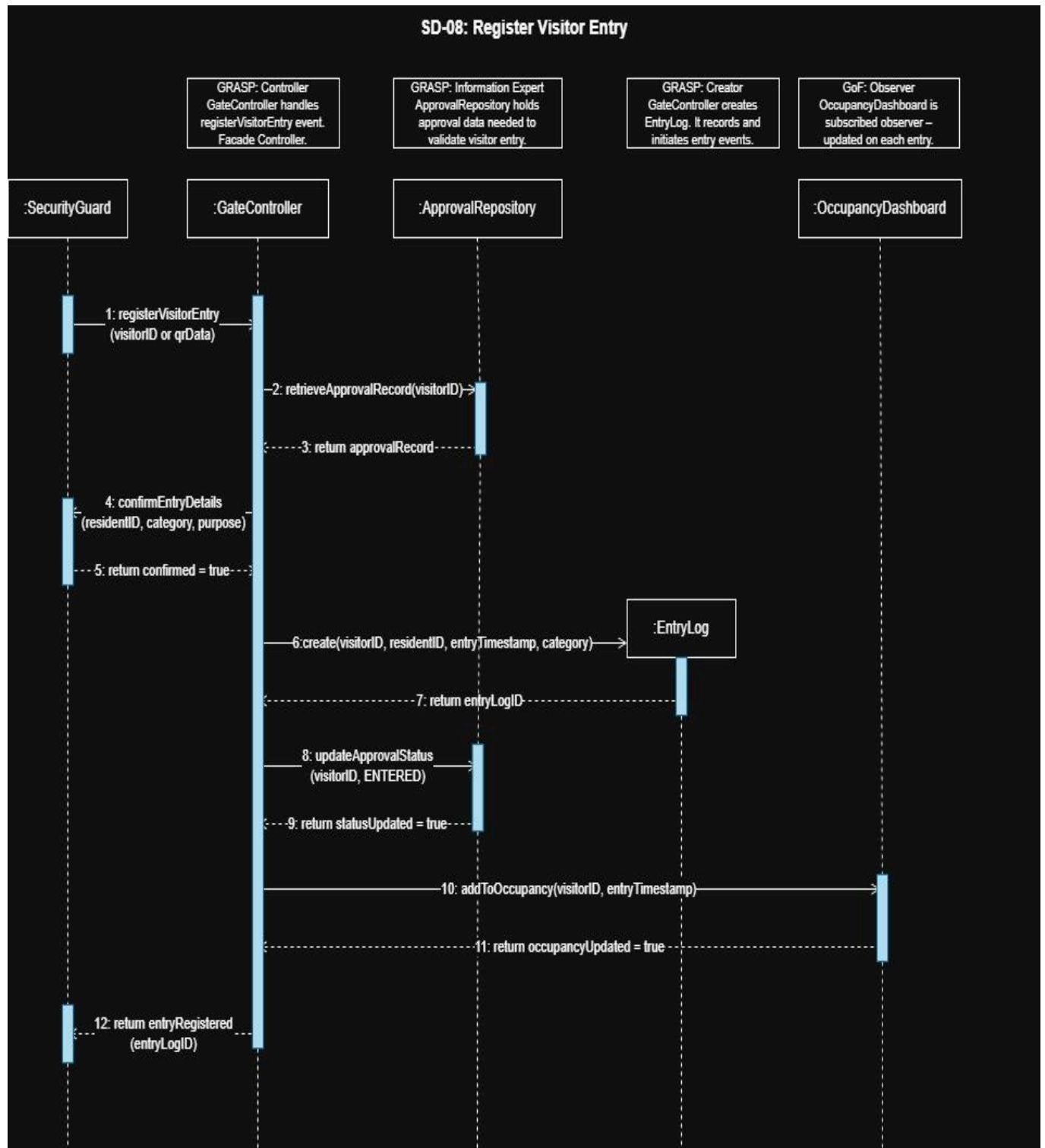
Use Case 6: View Visitor History



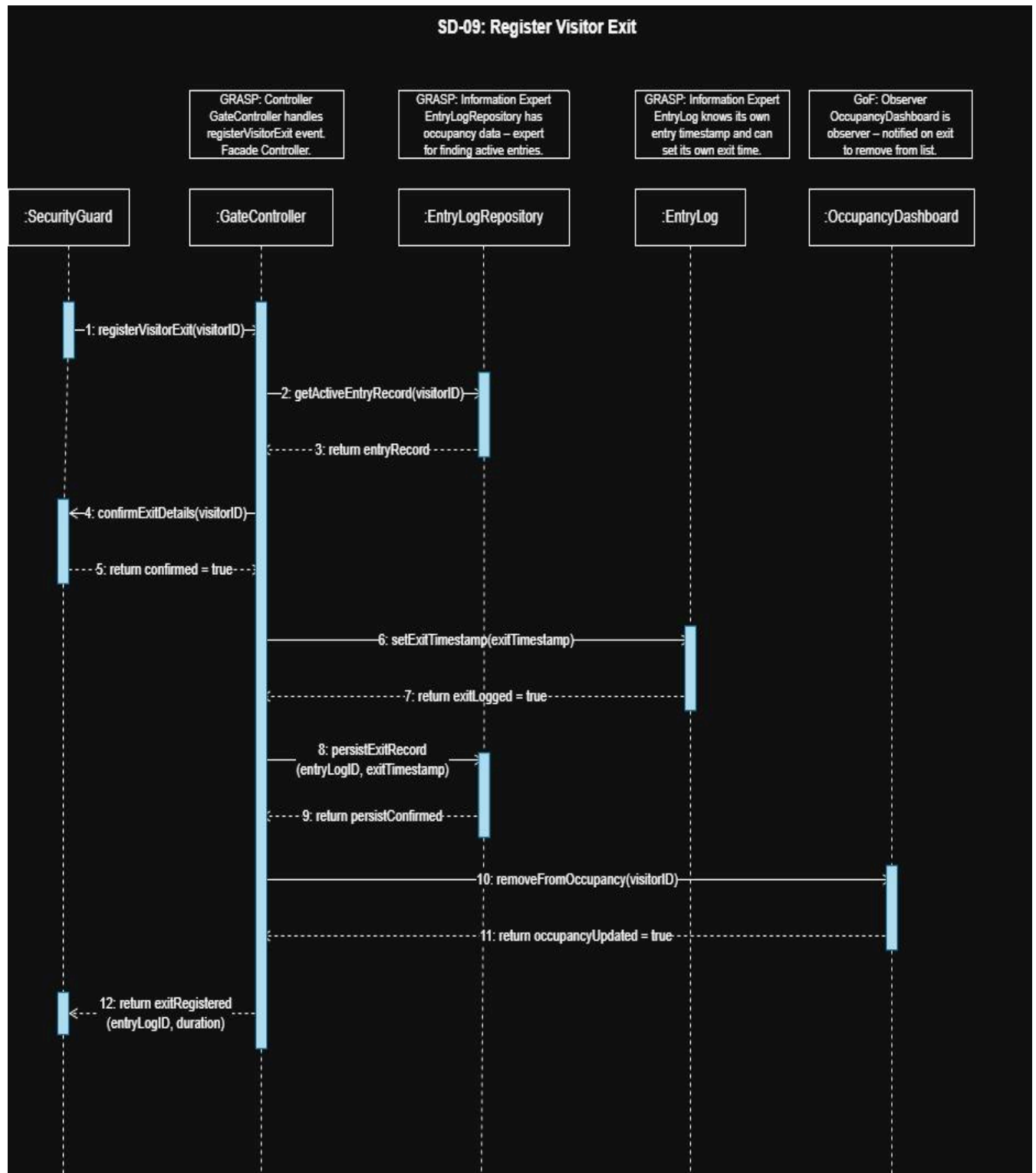
Use Case 7: Verify Visitor Approval Status



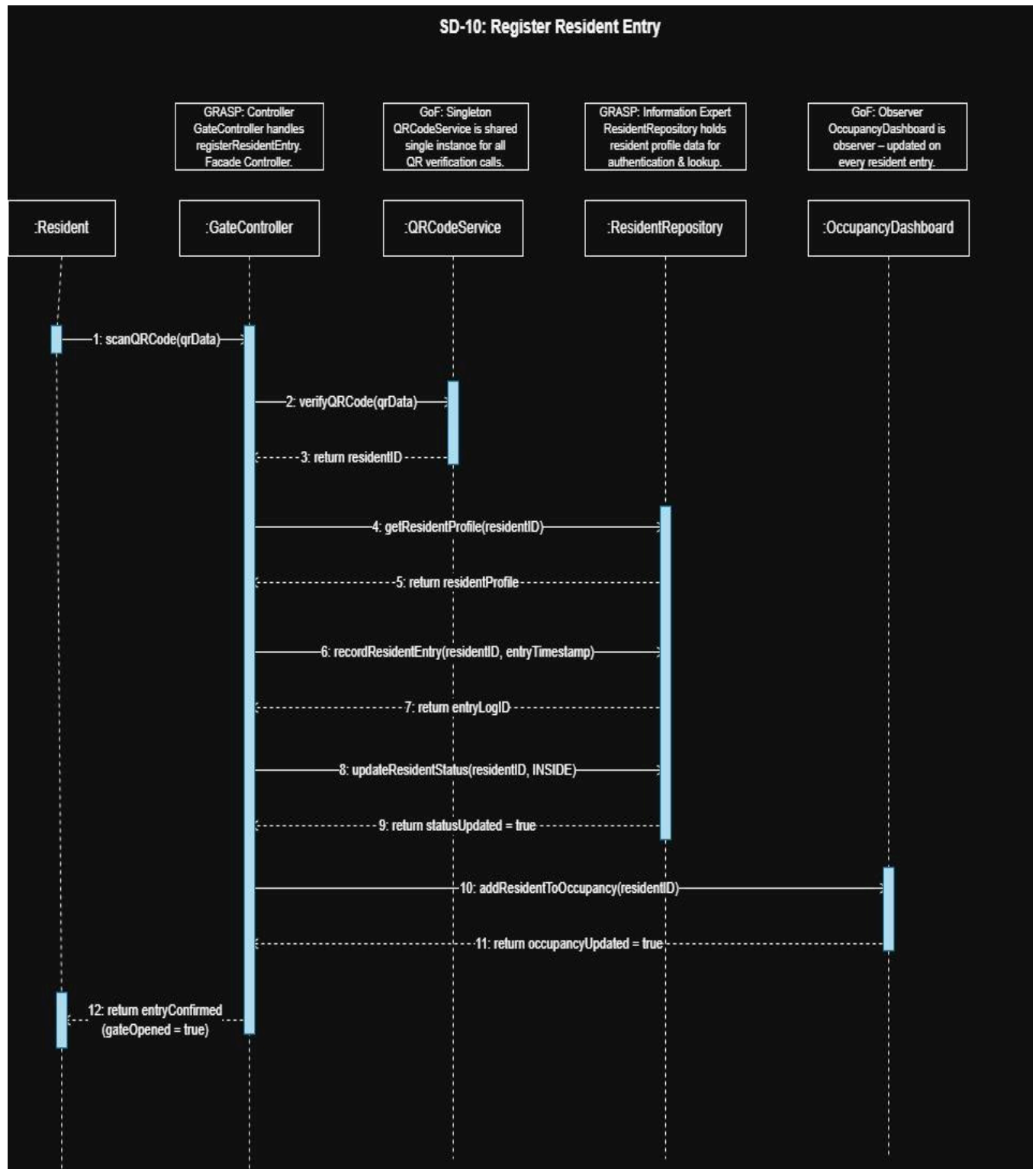
Use Case 8: Register Visitor Entry



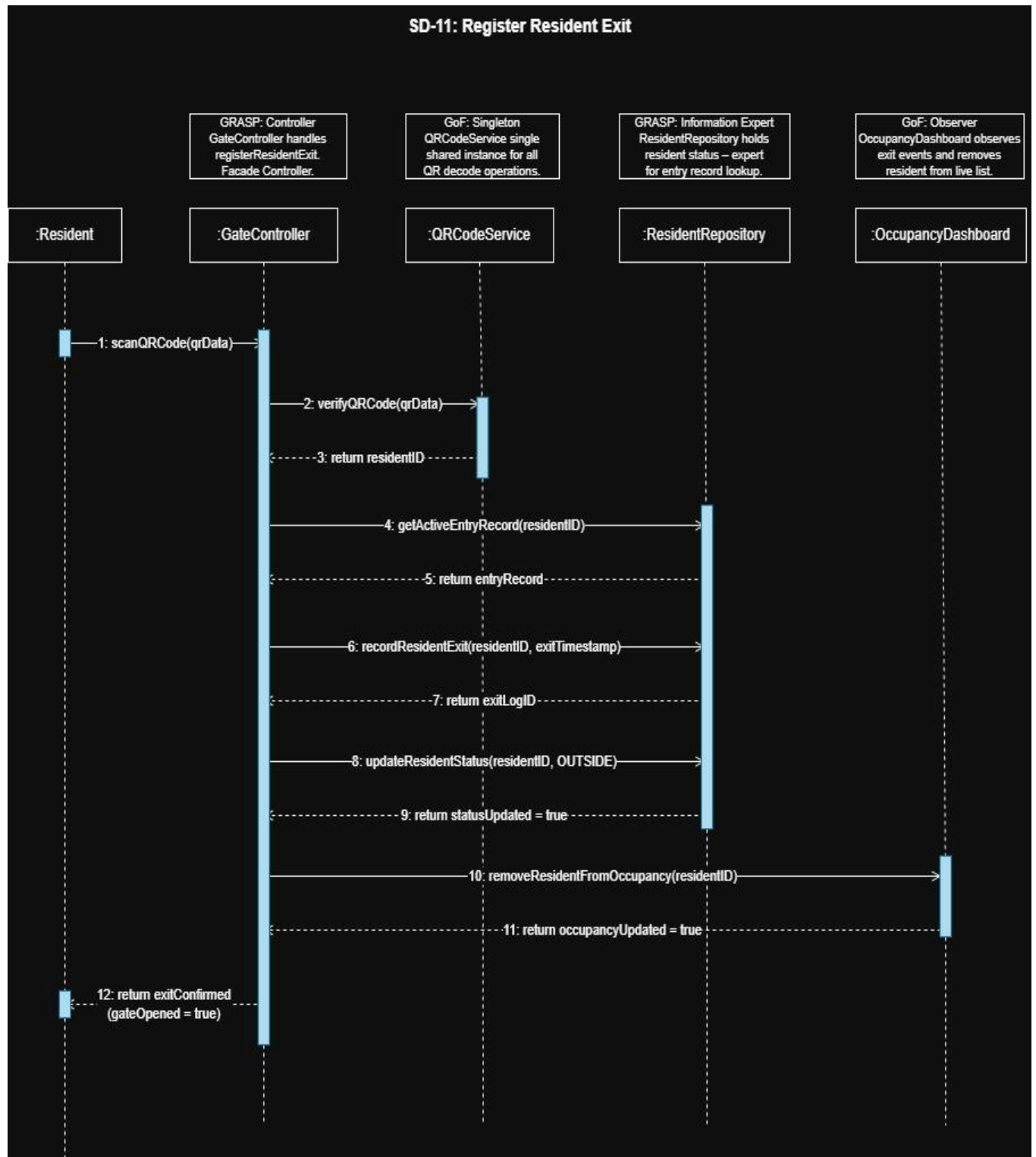
Use Case 9: Register Visitor Exit



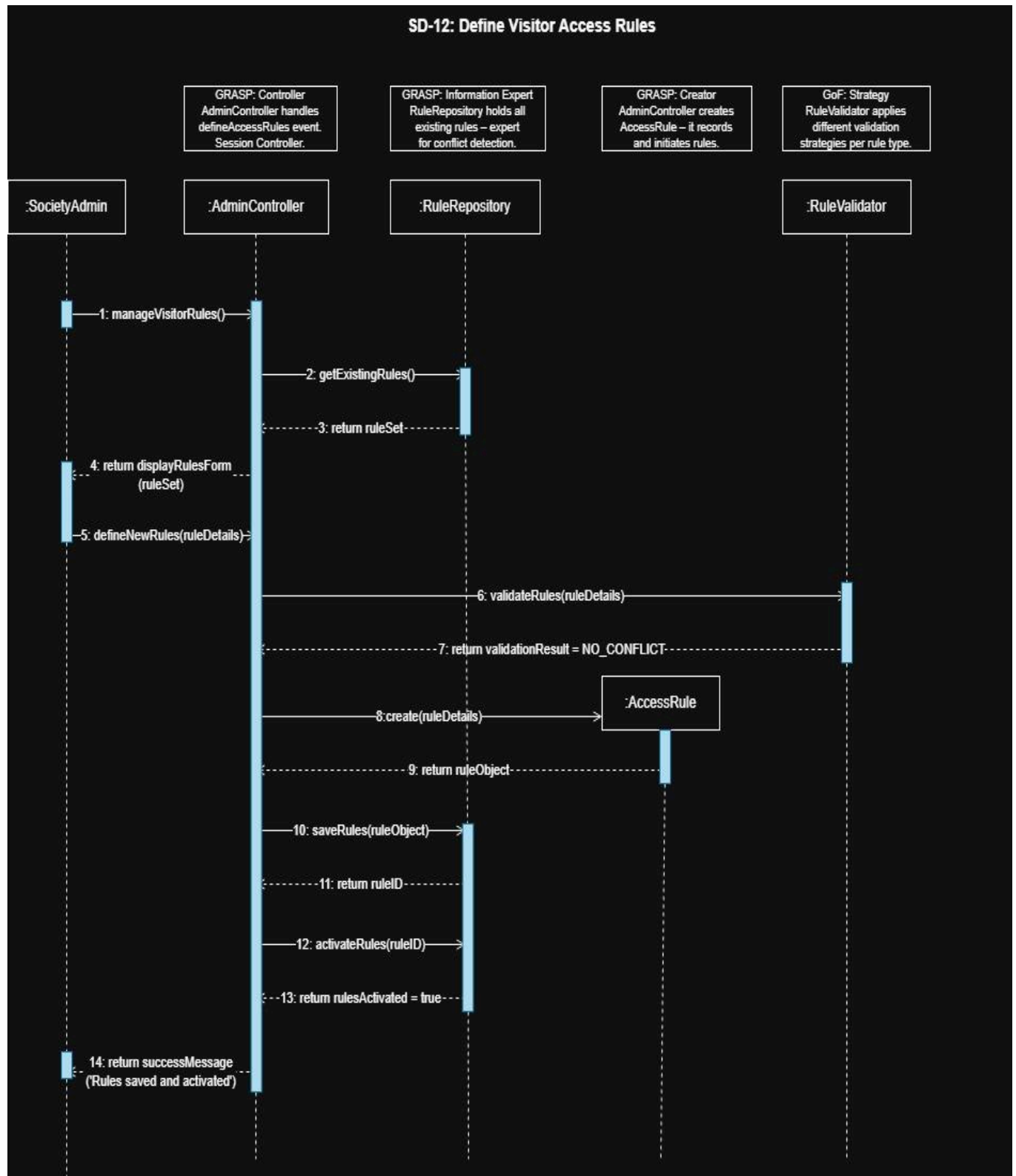
Use Case 10: Register Resident Entry



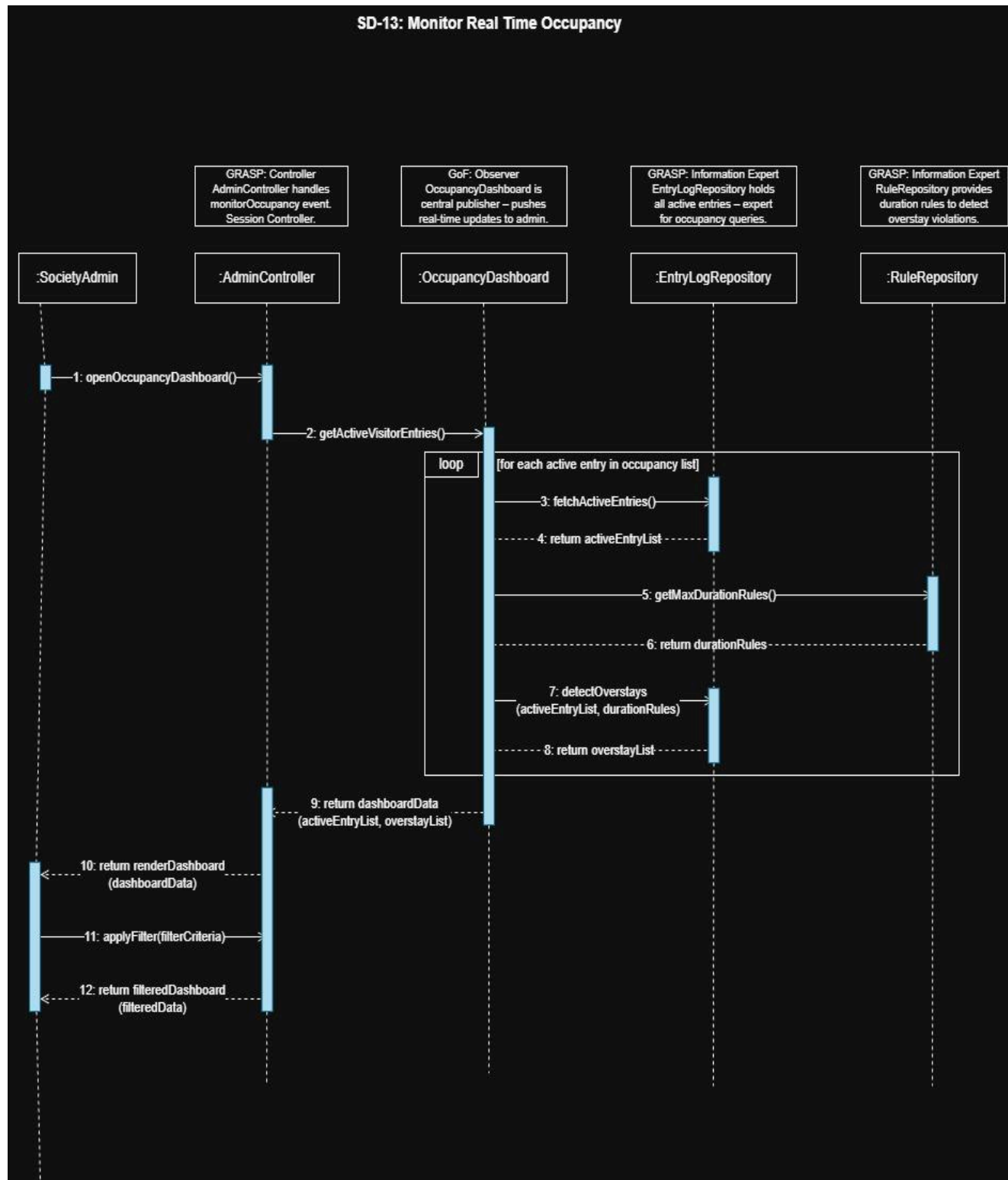
Use Case 11: Register Resident Exit



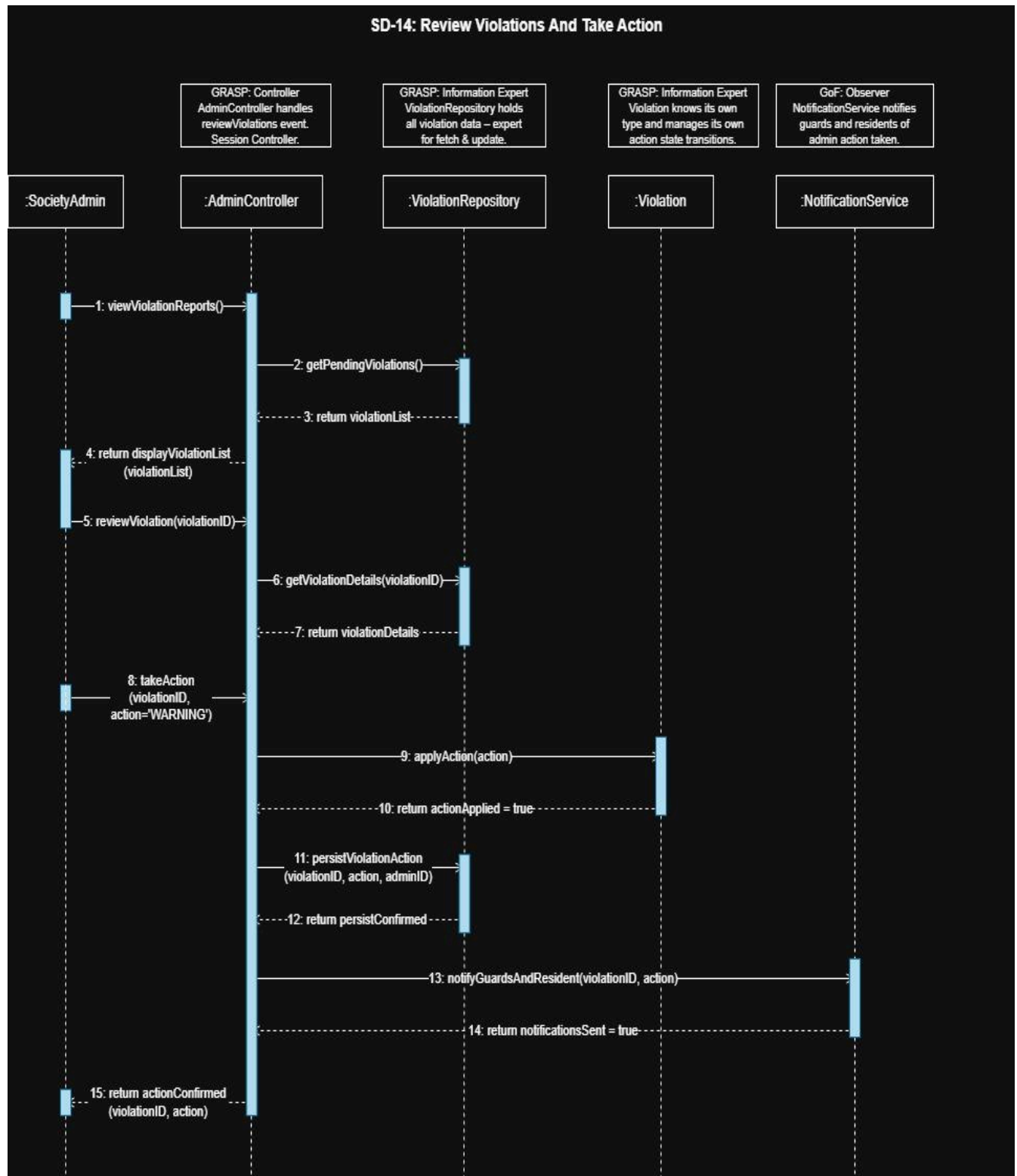
Use Case 12: Define Visitor Access Rule



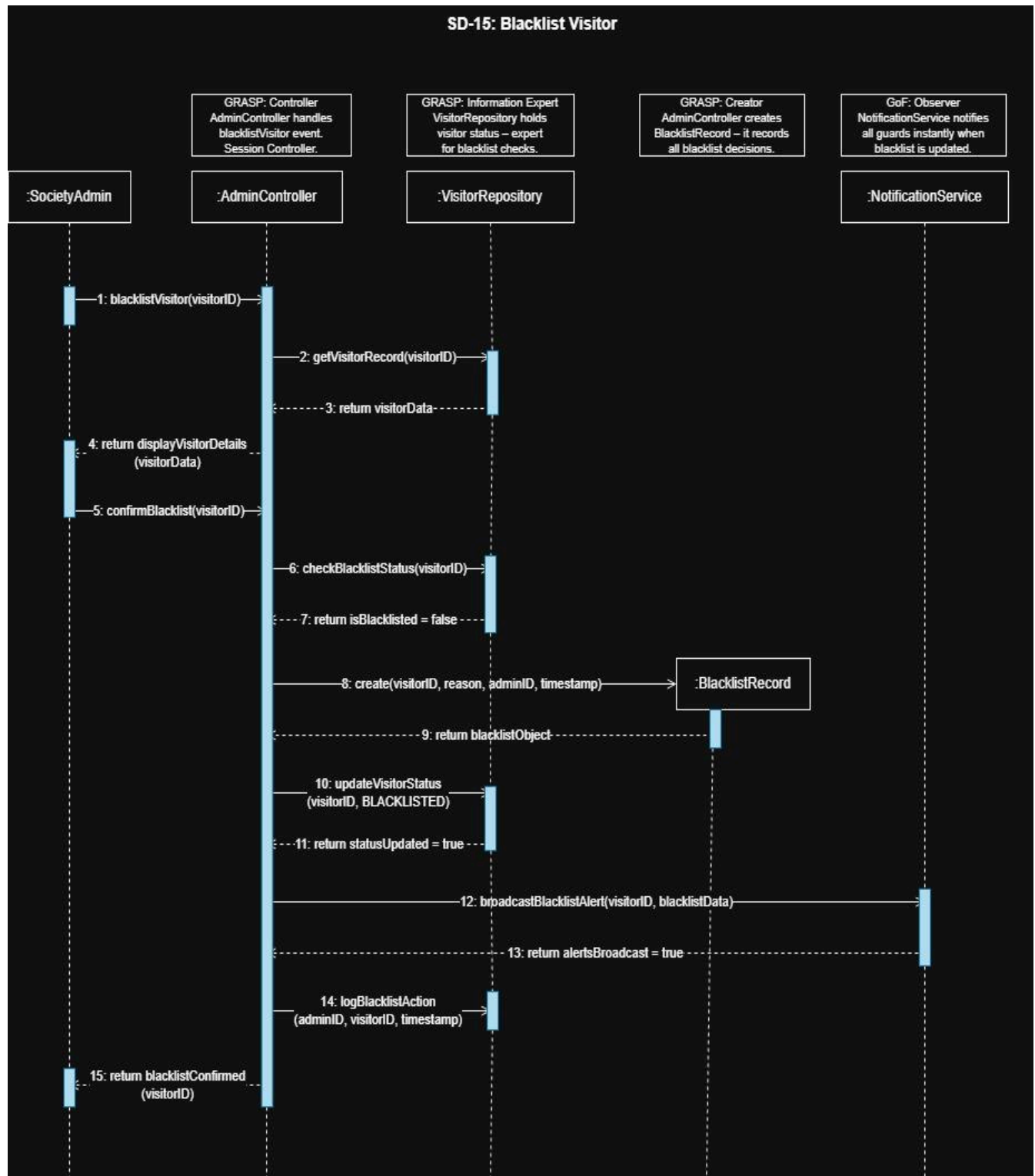
Use Case 13: Monitor Real Time Occupancy



UseCase 14: Review Violation and Take Action



Use Case 15: BlackList Visitor



Class Diagram:

